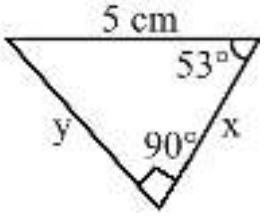
PHYSICS

1) $(0.996)^{1/4}$ = ... :-

(1) 0.992
(2) 1.002
(3) 0.999
(4) 1.001

2)

x, y and perimeter of the triangle are respectively :-



(1) 3 cm, 4 cm, 10 cm
(2) 4 cm, 3 cm, 12 cm
(3) 3 cm, 4 cm, 12cm
(4) 5 cm, 4 cm, 15 cm

3)

$(1 - \cos^2 15°) + (1 - \sin^2 15°)$ = ... :-

(1) 2
(2) 1
(3) –1
(4) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

4)

If $y = \frac{1}{x^{3/2}}$ then $\frac{dy}{dx}$ is :-

(1) $\frac{-3}{2}x^{-5/2}$
(2) $\frac{1}{2}x^{1/2}$
(3) $\frac{5}{2}x^{5/2}$

(4) $\frac{-1}{2}x^{-1/2}$

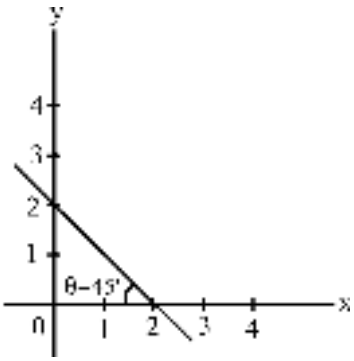
5) Value of sin 4° is approximately equal to :-

(1) 4

(2) $\frac{\pi}{45}$

(3) $\frac{45}{\pi}$

(4) 1

6) Which equation will represent to the graph 

(1) y = x – 2

(2) y = x + 2

(3) x = y + 2

(4) x = –y + 2

7) The slope of the tangent to the curve

y = ℓn (cosx) at x = $\frac{3\pi}{4}$ is :-

(1) 1

(2) –1

(3) ℓn $\sqrt{2}$

(4) $\frac{1}{\sqrt{2}}$

8)

If $y = 2x^2 - 8x$, then minimum value of y is :-

(1) – 8

(2) 3

(3) – 3

(4) 8

9) Find the value of $\int \left(\frac{a}{x^2} + \frac{b}{x}\right) dx$

{Where a & b are constant} :-

(1) $\frac{a}{x} + b \log_e.x + C$

(2) $-\frac{a}{x} + b \log_e.x + C$

(3) $a \log_e.x + \frac{b}{x} + C$

(4) $- a \log_e.x + \frac{b}{x} + C$

10) If $y = \sin(e^x)$, then $\frac{dy}{dx}$ = ... ?

(1) $\sin(e^{-x})$

(2) $\cos(e^x)$

(3) $e^x \times \sin(e^x)$

(4) $e^x \times \cos(e^x)$

11) A mettalic disc is being heated. Its area
A (in m^2) at any time t (in second) is given by $A = 5t^2 + 4t + 8$. Calculate the rate of increase in area at t = 3 sec :-

(1) 44

(2) 32

(3) 34

(4) 22

12) $\frac{d}{dx}(x^2e^x) =$

(1) $x^2e^x + e^xx^2$

(2) $x^2e^x + e^x2x$

(3) e^x2x

(4) $x^2e^x - e^x2x$

13) Integral of $\int_{-\pi/2}^{0} \sin x\, dx + \int_{0}^{\pi/2} \sin x\, dx$ is equal to :-

(1) 0

(2) 2

(3) –2

(4) 1

14) $\int_{3}^{5} \frac{1}{2x+3}\,dx$ is equal to :-

(1) $\ln\left(\frac{13}{9}\right)$

(2) $\frac{1}{2}\ln\left(\frac{13}{9}\right)$

(3) $\frac{1}{2}\ln\left(\frac{15}{9}\right)$

(4) None of these

15) Area bounded by curve y= sinx, and x-axis when x varies from 0 to π/2 is :-

(1) 1 Unit

(2) 2 Unit

(3) 3 Unit

(4) 0 Unit

16) $\int_{0}^{2\pi} (\cos^2\theta)\,d\theta$ will be equal to :-

(1) π

(2) 2π

(3) 0

(4) π/2

17)

$1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots\ldots \infty =$

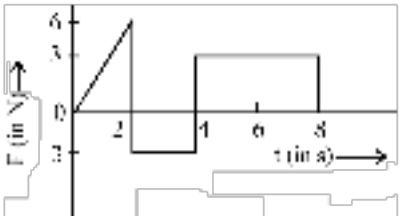
(1) $\frac{4}{3}$

(2) $\frac{3}{2}$

(3) $\frac{5}{2}$

(4) $\frac{13}{9}$

18) The force 'F' acting on a particle of mass 'm' is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from zero to 8 s is :-



(1) 24 Ns

(2) 20 Ns

(3) 12 Ns

(4) 6 Ns

19) Bullets of mass 50g are fired from a gun of mass 10 kg with velocity of 300 m/s. If 5 bullets are fired per second then what force a person will have to apply on the gun so that it does not recoil:-

(1) 75 N

(2) 150 N

(3) 225 N

(4) 300 N

20) When a force F is applied on a body of mass M, the acceleration produced is 'a'. If three forces equal to F and inclined to each other at 120° acts on the same body the acceleration produced will be :-

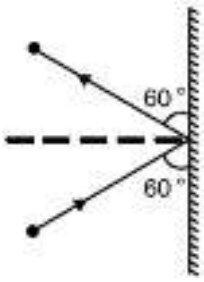
(1) $\sqrt{2}a$

(2) $\sqrt{3}a$

(3) 3a

(4) 0

21) A 3 kg ball strikes with a heavy rigid wall with a speed of 10 ms^{-1} at an angle of 60°. It gets reflected with the same speed and same angle as shown in the figure. If the ball is in contact with



the wall for 0.2 s, what is the average force exerted on the ball by the wall ?

(1) 150 N

(2) 0 N

(3) $150\sqrt{3}N$

(4) 55 N

22) A force applied on a body is represented as $\vec{F} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ N. Body accelerates with $1m/s^2$. What will be the mass of the body:-

(1) $10\sqrt{2}$ kg

(2) $2\sqrt{10}$ kg

(3) 10 kg

(4) 20 kg

23) A vehicle of mass 100 kg is moving with a velocity of 5 m/sec. To stop it in $\frac{1}{10}$ sec, the required force in opposite direction is :-

(1) 5000 N

(2) 500 N

(3) 50 N

(4) 1000 N

24) Given $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$. Which of the following statements is **INCORRECT** ?

(1) $\vec{A} \cdot \vec{B} = 50$

(2) $\frac{|\vec{A}|}{|\vec{B}|} = \frac{1}{2}$

(3) $|\vec{A}| = 5$

(4) $\vec{A} \cdot \vec{B} = 48$

25) The resultant of which pair cannot have a magnitude of 4N ?

(1) 2N and 3N

(2) 3N and 3N

(3) 2N and 6N

(4) 3N and 8N

26)

Two vector $\vec{a} = 3\hat{i} + 8\hat{j} - 2\hat{k}$ and $\vec{b} = 6\hat{i} + 16\hat{j} + x\hat{k}$ are such that the component of $\vec{b}$ perpendicular to $\vec{a}$ is zero. Then the value of x will be :-

(1) 8

(2) -4

(3) +4

(4) -8

27) The vector sum of two force is perpendicular to their vector differences. In that case, the forces :-

(1) Cannot be predicted

(2) Are perpendicular to each other

(3) Are equal to each other in magnitude

(4) Are not equal to each other in magnitude

28) A particle is moving with speed 6 m/s along the direction of $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$, then its velocity is :

(1) $(4\hat{i} + 2\hat{j} - 4\hat{k})$ unit

(2) $(4\hat{i} + 4\hat{j} - 2\hat{k})$ unit

(3) $(4\hat{i} + 4\hat{j} - 4\hat{k})$ unit

(4) $(2\hat{i} + 4\hat{j} - 2\hat{k})$ unit

29) $\vec{A} = 0.3\hat{i} + b\hat{j} - 0.4\hat{k}$ is an unit vector what is the value of b

(1) 0.25

(2) 0.75

(3) $\sqrt{0.75}$

(4) 0.93

30) The vector $\vec{P}$ makes 120° with the x-axis and the vector $\vec{Q}$ makes 30° with the y-axis. What is their resultant ?

(1) P + Q

(2) P – Q

(3) $\sqrt{P^2 + Q^2}$

(4) $\sqrt{P^2 - Q^2}$

31) Find value of $\left(2\vec{A} + 3\vec{B}\right) . \left(3\vec{A} - 2\vec{B}\right)$

(1) zero

(2) $6AB\cos\theta + 9A^2 - 9B^2$

(3) $6A^2 + 5AB\cos\theta - 6B^2$

(4) $5AB\cos\theta$

32) A vector of length □ is turned through the angle θ about its tail. What is the change in the position vector of its head :

(1) □ cos(θ/2)

(2) 2□ sin(θ/2)

(3) 2□ cos(θ/2)

(4) □ sin(θ/2)

33) With respect to a rectangular cartesian co-ordinate system three vectors are expressed as $\vec{a} = 4\hat{i} - \hat{j}$, $\vec{b} = -3\hat{i} + 2\hat{j}$, and $\vec{c} = -\hat{k}$ where $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along the x,y, z axes respectively. The unit vector along the direction of sum of these vectors is :-

(1) $\hat{r} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} - \hat{k})$

(2) $\hat{r} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j} - \hat{k})$

(3) $\hat{r} = \frac{1}{3}(\hat{i} - \hat{j} + \hat{k})$

(4) $\hat{r} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

34) If $\vec{a}_1$ and $\vec{a}_2$ are two non collinear unit vectors and $|\vec{a}_1 + \vec{a}_2| = \sqrt{3}$, then the value of $(\vec{a}_1 - \vec{a}_2) \cdot (2\vec{a}_1 + \vec{a}_2)$ is –

(1) 2

(2) $\frac{3}{2}$

(3) $\frac{1}{2}$

(4) 1

35) If $\hat{a}$ and $\hat{b}$ are non-collinear unit vectors and $\left|\hat{a} + \hat{b}\right| = \sqrt{3}$, then find the value of $(2\hat{a} - 5\hat{b}).(3\hat{a} + \hat{b})$.

(1) $\frac{41}{2}$

(2) $\frac{11}{2}$

(3) $-\frac{11}{2}$

(4) $-\frac{41}{2}$

36) Match the columns :-

	Column-I		**Column-II**
(A)	$\left(3\hat{i} + 2\hat{j} + 3\hat{k}\right) . \left(\hat{i} - \hat{j} + \hat{k}\right)$	(P)	0
(B)	$\left\|\left(\hat{i} + \hat{j} + \hat{k}\right) \times \left(2\hat{i} - \hat{j} + 2\hat{k}\right)\right\|$	(Q)	$3\sqrt{2}$
(C)	$\left(\hat{a} + \hat{b}\right) . \left(\hat{a} \times \hat{b}\right)$	(R)	4
(D)	$\left(\hat{a} + \hat{b}\right) . \left(\hat{a} - \hat{b}\right)$	(S)	2

(1) A – R, B – Q, C – P, D – S

(2) A – S, B – R, C – S, D – P

(3) A – R, B – Q, C – P, D – P

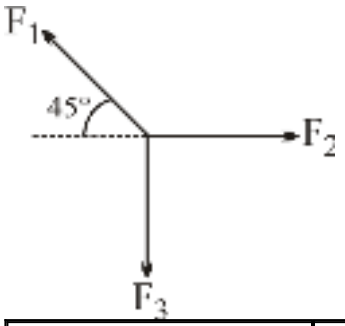
(4) A – R, B – S, C – P, D – P

37) Consider two vectors
$\vec{A}$ = (50N) north
$\vec{B}$ = (2m) west
The cross product $\vec{A} \times \vec{B}$ is :-

(1) (100 N-m) east
(2) (25 N-m) up
(3) (100 N-m) down
(4) (100 N-m) up

38)

Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are represented as shown. Each of them is of equal magnitude.



List-I (Combination)		List-II (Approximate Direction)	
(P)	$\vec{F}_1 + \vec{F}_2 + \vec{F}_3$	(1)	
(Q)	$\vec{F}_1 - \vec{F}_2 + \vec{F}_3$	(2)	
(R)	$\vec{F}_1 - \vec{F}_2 - \vec{F}_3$	(3)	
(S)	$\vec{F}_2 - \vec{F}_1 - \vec{F}_3$	(4)	

(1) P-1, Q-2, R-3, S-4
(2) P-2, Q-1, R-4, S-3
(3) P-2, Q-3, R-1, S-4
(4) P-4, Q-1, R-2, S-3

39)

What is value of $(\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{A})$?

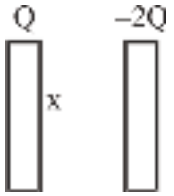
(1) AB sin θ
(2) A^2B^2 sin θ
(3) A^2B^2 sin θ
(4) $-A^2B^2 \sin^2 \theta$

40) **Statement-I :** $\vec{A} \times \vec{B}$ is perpendicular to $\vec{A} + \vec{B}$ and $\vec{A} - \vec{B}$.
Statement-II : $\vec{A} \times \vec{B}$ and $\vec{B} \times \vec{A}$ are anti parallel and non coplanar.

(1) Statement-I is correct but statement-II is incorrect.
(2) Statement-I is incorrect but statement-II is correct.
(3) Statement-I and II both are correct.
(4) Both statement-I and II are incorrect.

41) Which value of charge is not possible :-

(1) 6.4×10^{-18}C
(2) 8.0×10^{-17} C
(3) 9.6×10^{-19} C
(4) 3.2×10^{-20} C

42) Find the value of x for large metallic plates :- 

(1) –Q
(2) 2Q
(3) $\frac{+3Q}{2}$
(4) 3Q

43) A & B are two identical metallic spheres, charge on them is 7μC and 1μC. If both are connected by a wire, then amount of charge which flows from A to B:-

(1) 8μC
(2) 4μC
(3) 3μC
(4) 1.5μC

44) Find correct statements :-
(A) Charge is invariant while mass is not.
(B) For an isolated system charge is always conserved while mass is not always conserved.
(C) Charge of proton is equals to sum of charge of 2 up quark and 1 down quark.
(D) Repulsion is the true test of electrification.

(1) A, B, C, D
(2) A, B, D
(3) A, C, D
(4) B, D

45) Two point charges + 2C and +6C repel each other with a force of 12N. If a charge of –4C is

given to each of these charges, the force now is :- (Keeping distance between charges same)

(1) 4 N (repulsive)
(2) 4 N (attractive)
(3) 12 N (attractive)
(4) 8 N (repulsive)

CHEMISTRY

1) For the reaction; A + 2B → C, 5 mole of A and 8 mole of B will produce :

(1) 5 mole of C
(2) 4 mole of C
(3) 8 mole of C
(4) 13 mole of C

2) Equal masses of O_2 and N_2 are reacted according to the equation : $O_2 + N_2 \rightarrow 2NO$. Which statement is true ?

(1) O_2 is the limiting reagent and N_2 is present in excess
(2) N_2 is the limiting reagent and O_2 is present in excess.
(3) All of O_2 and N_2 react and neither is in excess.
(4) Nothing can be said about the limiting reagent

3) What mass of Al must taken so as to produce Al_2O_3 by reacting it with 32 g O_2 :- $4Al + 3O_2 \rightarrow 2Al_2O_3$

(1) 27 g
(2) 96 g
(3) 36 g
(4) 54 g

4) 4 moles of H_2 gas are ignited with 4 moles of O_2 gas. The weight of water formed is $2H_2 + O_2 \rightarrow 2H_2O$

(1) 36 g
(2) 18 g
(3) 72 g
(4) 9 g

5) What weight of 95% $CaCO_3$ will be required to produce 11.2 litre CO_2 :-

(1) 50 g
(2) 100 g
(3) 52.6 g

(4) 25 g

6) Different proportions of oxygen in the various oxides of nitrogen prove the law of -

(1) Gay Lussac's Law
(2) Multiple proportion
(3) Constant proportion
(4) Conservation of mass

7) Which of the following pairs of compounds does not follow law of multiple proportions ?

(1) CO, CO_2
(2) H_2O, H_2O_2
(3) CaO, MgO
(4) N_2O, N_2O_3

8) How many electrons are present in 2×10^{-3} mol of ${}_{8}^{18}O^{2-}$?

(1) 1.2×10^{21}
(2) 9.6×10^{21}
(3) 1.2×10^{22}
(4) 1.9×10^{22}

9) Number of molecules in 1 litre of oxygen at NTP is :-

(1) $\frac{6.02 \times 10^{23}}{32}$
(2) 32×22.4
(3) $\frac{6.02 \times 10^{23}}{22.4}$
(4) $\frac{32}{22.4}$

10) Arrange the following in increasing order of mass ?
(I) 0.5 mole of O_3
(II) 0.5 gm atom of oxygen
(III) 3.011×10^{23} molecules of O_2
(IV) 5.6 litre of CO_2(g) at 1 atm and 0°C

(1) II < IV < III < I
(2) II < I < IV < III
(3) IV < II < III < I
(4) I < II < III < IV

11) 30mL of a gaseous hydrocarbon requires 90 mL of oxygen for complete combustion and produces 60mL of CO_2, then the molecular formula of hydrocarbon is :-

(1) C_2H_2

(2) C_3H_8

(3) C_4H_{10}

(4) C_2H_4

12) The volume occupied by 1.8 g of water vapour at 374 °C and 1 bar pressure will be :-
[Use R = 0.083 bar L $K^{-1}mol^{-1}$]

(1) 96.66 L

(2) 55.87 L

(3) 3.10 L

(4) 5.37 L

13) In the reaction $N_2 + 3H_2 \rightarrow 2\ NH_3$, ratio by volume of N_2, H_2 and NH_3 is 1 : 3 : 2. This illustrates law of -

(1) Difinite proportion

(2) Multiple proportion

(3) Law of conservation of mass

(4) Gaseous volumes

14) A gas mixture of 3 L of propane and butane on complete combustion at 25°C produces 10 L of CO_2. Initial composition of the propane & butane in the gas mixture is –

(1) 66.67%, 33.33%

(2) 33.33%, 66.67%

(3) 50%, 50%

(4) 60%, 40%

15) 90 mL of pure dry O_2 is subjected to silent electric discharge. If only 10% of it is converted to O_3, volume of the mixture of gases (O_2 and O_3) after the reaction will be ------ and after passing through turpentine oil will be .....

(1) 84 mL and 78 mL

(2) 81 mL and 87 mL

(3) 78 mL and 84 mL

(4) 87 mL and 81 mL

16) Which of the following relation is incorrect :-

(1) 1 mol N^{-3} ions have 10 mol electrons

(2) 1 mol O^{-2} ions have 6.023×10^{22} electrons

(3) 1 mole CH_4 has 10 mol protons

(4) 1 mole water has 10 mol protons

17) What will be the mass of one Al atom in gram:-

(1) 1.0×10^{-24} g

(2) 44.82×10^{-24} g

(3) 22.41×10^{-24} g

(4) 80×10^{-24} g

18) One mole of potassium chlorate ($KClO_3$) is thermally decomposed. How many mole of oxygen will formed :-

(1) 1

(2) 1.5

(3) 2

(4) 3

19) Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R) **Assertion (A)**: Both 106 g of sodium carbonate and 12 g of carbon have same number of carbon atoms.
Reason (R): Both contains 1 g atom of carbon which contains 6.02×10^{23} carbon atoms.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are true and **(R)** is not the correct explanation of **(A)**

(3) **(A)** is true but **(R)** is false

(4) **(A)** is false but **(R)** is true

20) An element X reacts with oxygen to form a compound X_2O_3. If the atomic mass of X is 91.5. The equivalent mass of X is:-

(1) 30.5

(2) 45.75

(3) 61

(4) 91.5

21) What is the percentage purity of a sample of $CaCO_3$ if 150 g of the sample gives out 44g of CO_2 gas on heating :-

(1) 33.33%

(2) 50%

(3) 66.67%

(4) 75%

22) Which of the following pair have the same number of atoms ?

(1) 16 g of O_2(g) and 44 g of CO_2(g)

(2) 44 g of CO_2(g) and 80 g of SO_3(g)

(3) 28 g of N_2(g) and 32 g of O_2(g)

(4) 34 g of NH_3(g) and 88 g of CO_2(g)

23) 200 mL of 0.1 M H_3PO_2 solution on complete neutralisation requires ..... mL of 0.5 M NaOH solution.

(1) 20 mL
(2) 40 mL
(3) 120 mL
(4) 80 mL

24) If M^{+3} has configuration $[Ar]3d^{10}$ then M belongs to :-

(1) s-block
(2) p-block
(3) d-block
(4) f-block

25) The general configuration of d-block elements is :-

(1) $ns^2\ np^3$
(2) $ns^1\ (n-1)d^{0-1}$
(3) $ns^{0-2}\ (n-1)d^{1-10}$
(4) $ns^{1-2}\ (n-1)d^{1-10}\ (n-2)f^3$

26) Atomic number of Ag is 47. In the same group, the atomic numbers of elements placed above and below Ag will be:

(1) 37, 67
(2) 29,79
(3) 39, 69
(4) 29, 65

27) Give correct IUPAC symbol for last chalcogens :-

(1) Uus
(2) Uuh
(3) Uuo
(4) Uup

28) In which of the following compounds chromium (Cr) shows minimum radius ?

(1) CrO
(2) Cr_2O_3
(3) CrO_3
(4) None

29) Which of the following process involves maximum energy ?

(1) $K^{+} \rightarrow K^{+2} + e^{\ominus}$

(2) $Na^{+} \rightarrow Na^{+2} + e^{\ominus}$

(3) $Li^{+} \rightarrow Li^{+2} + e^{\ominus}$

(4) $Be^{+2} \rightarrow Be^{+3} + e^{\ominus}$

30) For which of the following Z_{eff} is equal to Z :-

(1) H

(2) He

(3) Na

(4) Li

31) Correct order of shielding effect is :-

(1) p > d > f

(2) d > p > s

(3) p > f > s

(4) s > d > p

32) In which of the following pairs both the species have nearly same size :-

(1) Li^{+}, Mg^{2+}

(2) Mo, W

(3) Co, Ni

(4) All of these

33)

Lanthanoid contraction will not have any influence when we compare size of :-

(1) Zr & Hf

(2) Y & La

(3) Mo & W

(4) Tc & Re

34) Which of the following configuration will have least I.P. value ?

(1) $1s^2 2s^2 2p^6 3s^2 3p^6$

(2) $1s^2 2s^2 2p^6 3s^2 3p^5$

(3) $1s^2 2s^2 2p^6 3s^2$

(4) $1s^2 2s^2 2p^6 3s^2 3p^1$

35) Electronic configuration of valence shell of four elements A, B, C and D are :-
(A) $ns^2\ np^1$ (B) $ns^2\ np^2$
(C) $ns^2\ np^3$ (D) $ns^2\ np^4$

Correct increasing order of their second ionisation energies is :-

(1) A < B < C < D
(2) A < B < D < C
(3) B < A < D < C
(4) B < A < C < D

36) IE values for Li and K are 5.4 and 4.3 eV respectively. IE of Na will be :

(1) 9.7 eV
(2) 1.1 eV
(3) 4.9 eV
(4) 6.2 eV

37) Which of the following is best reducing agent in gaseous state ?

(1) Mg
(2) Be
(3) Na
(4) K

38) The correct order of electron affinity of Halogen family is :-

(1) F > Cl > Br > I
(2) Cl > F > Br > I
(3) I > Br > F > Cl
(4) None of these

39) What is the value of electron gain enthalpy of Na^+ if IE_1 of Na = 5.1 eV :-

(1) +0.2 eV
(2) –5.1 eV
(3) –10.2 eV
(4) +2.55 eV

40) In the formation of a chloride ion, from an isolated gaseous chlorine atom, 3.8 eV energy is released, which would be equal to :-

(1) Electron affinity of Cl^-
(2) Ionisation potential of Cl
(3) Electronegativity of Cl
(4) Ionisation potential of Cl^-

41) Among the following element which has highest electron affinity :

(1) Si

(2) P

(3) S

(4) Cl

42) Which of the following bond is maximum polar?

(1) H–Cl

(2) H–Br

(3) H–F

(4) H–I

43) Which among the given formula of the compounds is not the correct representation ?

(1) OF_2

(2) OBr_2

(3) H_2O

(4) N_2O

44) Which of the following strongly absorbs CO_2 ?

(1) K_2O

(2) BeO

(3) P_4O_{10}

(4) Conc. H_2SO_4

45) The correct order of acidic strength is :-

(1) $Cl_2O_7 > SO_3 > P_4O_{10}$

(2) $CO_2 > N_2O_5 > SO_3$

(3) $Na_2O > MgO > Al_2O_3$

(4) $K_2O > CaO > MgO$

BIOLOGY

1) Read the statements and select the correct option.
Statement A: Cytoplasm is the main arena of cellular activities in only plant cells.
Statement B: Ribosomes are non-membrane bound organelles.

(1) Only statement A is correct

(2) Both statements A and B are correct

(3) Only statement B is correct

(4) Both statements A and B are incorrect

2) The centromere is present at the terminal end of the chromosome in

(1) Metacentric chromosome

(2) Acrocentric chromosome

(3) Telocentric chromosome

(4) Sub-metacentric chromosome

3) Plastid which is specific for storage of oils and fats is :-

(1) Amyloplast

(2) Aleuroplast

(3) Elaioplast

(4) Chromoplast

4) Which of the following statement is not related with rough endoplasmic reticulum ?
(A) Actively involved in protein synthesis & secretion
(B) Act as major site for synthesis of lipid.
(C) Involved in lipid like steroidal hormone synthesis in animal cells.
Option

(1) A & B only

(2) B & C only

(3) A & C only

(4) A, B & C only

5) Which statement accurately describes the structure of mitochondria?

(1) They are single-membrane bound organelles with a dense Matrix.

(2) Mitochondria possess a single ds-circular DNA molecule, a few RNA molecules, and ribosomes.

(3) The outer membrane of mitochondria forms infoldings called cristae.

(4) Mitochondria primarily store carbohydrates and proteins.

6) Secondary cell wall is formed

(1) Outside the primary cell wall

(2) Inside the cell membrane

(3) Inside the plasmodesmata

(4) Inside the primary cell wall

7) **Assertion :** Cillia and flagella are hair like out growth of cell membrane.
Reason : Cillia are small structures which work like oars causing the movement of cell.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

8) Central tubules are connected to one of the tubules of each peripheral doublet by a __A__ and

doublet show a pair of arm which made up of __B__.
Select correct option to fill A and B.

(1) A-Central bridge, B-Actin protein

(2) A-Central bridge, B-Mayosin protein

(3) A-Radial spoke, B-Dynein protein

(4) A-Radial spoke, B-Tubulin protein

9) What will be arrangement of microtubules in axoneme of eukaryotic flagella :-

(1) 0 + 2

(2) 4 + 4

(3) 9 + 2

(4) 9 + 0

10) **Statement-1 :** Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.
Statement-2 : Nucleolus is a site for active ribosomal RNA synthesis.

(1) Both the Statements are incorrect.

(2) Both the Statements are correct.

(3) Statement 1 is correct but Statement 2 is incorrect.

(4) Statement 1 is incorrect but Statement 2 is correct.

11) Which of the following are not present in higher plant cell but present in animal cell ?

(1) Ribosome

(2) Centrosome

(3) Chloroplast

(4) Mitochondria

12) Svedberg's unit or Sedimentation co-efficient is indirectly a measure of ___(a) and ___(b) ___.

(1) (a) Size, (b) Density

(2) (a) Density, (b) Mass

(3) (a) Length, (b) Size

(4) (a) Volume, (b) Size

13) **Statement-I :** The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes.
Statement-II : In Amoeba the contractile vacuole is important for osmoregulation and excretion.
Select correct option :

(1) Statement-I and Statement-II both are correct.

(2) Statement-I and Statement-II both are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

14) Mitochondria are the site of ___A___ and mitochondria divide by ___B___.
Select correct option for A and B :-

(1) A - Anaerobic respiration, B - Meiosis

(2) A - Aerobic respiration, B - Fission

(3) A - Anaerobic respiration, B - Mitosis

(4) A - Aerobic respiration, B - Meiosis

15) (I) Ribosomes first observed by __A__.
(II) Ribosomes are two types __B__.
(III) Number of Chloroplast in Chlamydomonas is __C__ per cell.
(IV) The inner compartment of mitochondria is filled with a dense homogeneous substance called the __D__.
Select correct option for A, B, C and D :-

(1) A - Robert Brown, B - 90 s and 65 s,
C - 11, D - Stroma

(2) A - George Palade, B - 70 s and 80 s,
C - 1, D - Matrix

(3) A - Robert Hook, B - 40 s and 25 s,
C - 11, D - Matrix

(4) A - Flemming, B - 15 s and 21 s,
C - 10, D - Stroma

16) The most abundant lipid in the cell membrane is :

(1) Cutin

(2) Cholesterol

(3) Steroid

(4) Phospholipid/phosphoglycerides

17)

Which of the following is common to facilitated diffussion and active transport ?

(1) Energy requirement

(2) Occurs along the concentration gradient

(3) Occurs against the concentration gradient

(4) Requirement of carrier protein.

18) Mark the incorrect match for transport of molecules across the membrane :-

(1) Neutral solute – simple diffusion

(2) Water – osmosis

(3) Polar molecules - facilitated diffusion

(4) ATP utilised – passive transport

19) Cytoplasmic connections, between two plant cell of multicellular plant is called as :-

(1) Tight junction
(2) Plasmodesmata
(3) Synapse
(4) Pits

20) Which cell wall is capable of growth and gradually diminishes as cell matures :-

(1) Primary cell wall
(2) Secondary cell wall
(3) Tertiary cell wall
(4) Primary and secondary cell wall

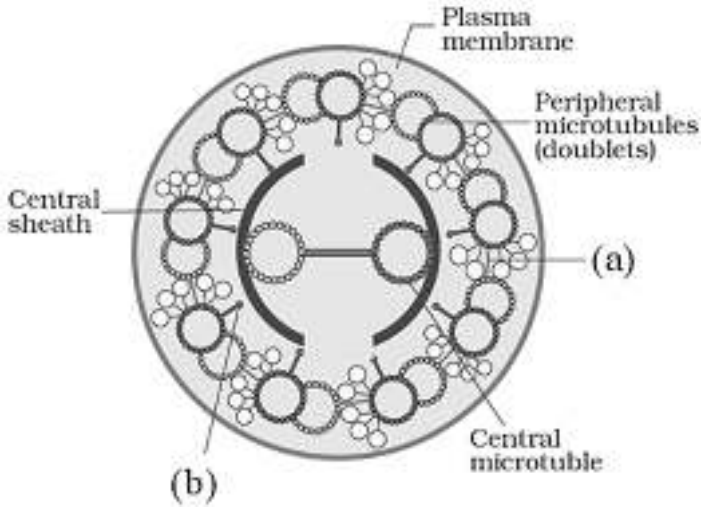
21) During different stages of cell division, cell show _____ in place of the nucleus. A single human cell has approximately _____ meter long thread of DNA distributed among its _____ chromosomes :

(1) Chromatin, 46, 5
(2) Chromosomes, 5, 46
(3) Chromosomes, 2, 46
(4) Nucleolus, 2, 23

22) Mark the mismatched pair :

(1)	Plastids	Found in plant cells and in euglenoids
(2)	Golgi apparatus	Site of formation of glycoproteins and glycolipids
(3)	Nucleus	During interphase it has highly extended and elaborate nucleoprotein fibers
(4)	Centriole	Microtubular organisation is referred to as the 9 + 2

(1) 1
(2) 2
(3) 3
(4) 4



23)
Identify (a) and (b) in given diagram :-

(1) (a) interdoublet bridge, (b) radial spoke
(2) (a) intertriplet bridge, (b) radial spoke
(3) (a) centriole, (b) intersinglet bridge
(4) (a) centrosome, (b) axoneme

24) Algae cell wall is made up of :

(1) Cellulose, hemicellulose and proteins
(2) Cellulose, galactans, mannans & $CaCO_3$ like minerals
(3) Cellulose, pectins & proteins only
(4) Only cellulose

25) Stroma of the chloroplast have enzymes for synthesis of :

(1) Carbohydrates
(2) Proteins
(3) Vitamins
(4) Both carbohydrates and proteins

26) Plastid differs from mitochondria on the basis of one of the following features. Mark the right answer :

(1) Presence of two layers of membrane
(2) Presence of ribosome
(3) Presence of chlorophyll
(4) Presence of DNA

27) When a chromosome in a stage of cell division appears 'V' shaped it is :

(1) Prophase
(2) Metaphase
(3) Telophase

(4) Anaphase

28) Match the following :

	List-I		**List-II**
(A)	Cristae	(I)	Infoldings of plasma membrane in Bacteria
(B)	Cisternae	(II)	Disc-shaped sacs in Golgi apparatus
(C)	Mesosomes	(III)	Infolding in Mitochondria
(D)	Thylakoids	(IV)	Flat membranous sacs in stroma

(1) A-I, B-II, C-III, D-IV

(2) A-III, B-IV, C-I, D-II

(3) A-II, B-III, C-I, D-IV

(4) A-III, B-II, C-I, D-IV

29) Identify the figure A, B and C given below :- 

	A	**B**	**C**
(1)	Mitochondria	Chloroplast	Golgi apparatus
(2)	Chloroplast	Ribosome	Lysosomes
(3)	Ribosome	Mitochondria	Chloroplast
(4)	Golgi apparatus	Ribosome	Mitochondria

(1) 1

(2) 2

(3) 3

(4) 4

30) Identify the set of correct statements :-
(A) The number of mitochondria per cell is fixed.
(B) Each mitochondria is a double membrane bound structure.
(C) The inner mitochondrial membrane forms a number of infolding towards the matrix.
(D) The chloroplast contain chlorophyll and carotenoids pigments which are responsible for trapping light energy.
(E) The prokaryotic ribosomes are 70S while the eukaryotic ribosomes are 800S.
Choose the correct answer from the options given below :-

(1) A, B and E only

(2) A, C and D only

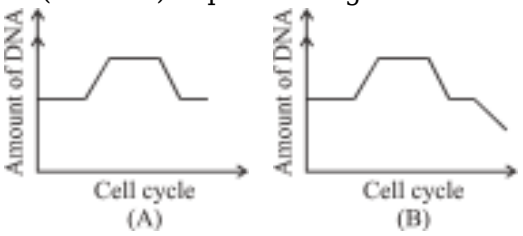
(3) B, C and D only

(4) C, D and E only

31) **Statement-I :-** The mitochondria divide by fission.
Statement-II :- Mitochondria are the sites of aerobic respiration.

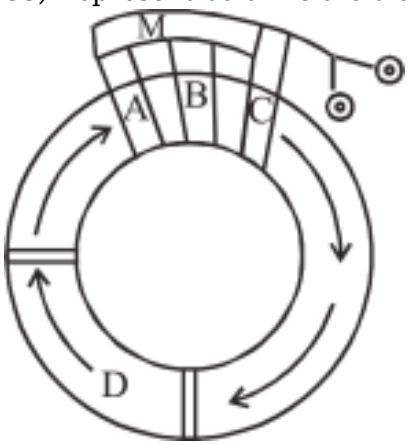
(1) Statement-I and statement-II both are false.
(2) Statement-I and statement-II both are true.
(3) Statement-I is true but statement-II is false.
(4) Statement-I is false but statement-II is true.

32) The following graphs (A and B) depicts changes in DNA Content during cell cycle. Identify the cell division A and B :-



(1) A → Meiosis–II, B → Meiosis–I
(2) A → Mitosis, B → Meiosis–I
(3) A → Meiosis–I, B → Mitosis
(4) A → Mitosis, B → Meiosis

33) Represent below is the diagram of cell–cycle. Which one of the following is correct for C ?



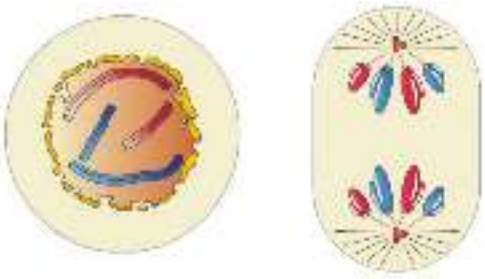
(1) Telophase
(2) Anaphase
(3) Cytokinesis
(4) G_2 phase

34) Some events of mitotic cell division are given below in unordered manner -
(i) Condensation of chromatin is completed
(ii) Chromosome lose their identity as discrete elements
(iii) Sister chromatid separate
(iv) Initiation of the assembly of mitotic spindle between pair of centrioles
The correct order for steps of cell division is :-

(1) i → iv → iii → ii
(2) iv → i → iii → ii

(3) ii → i → iv → iii

(4) iv → iii → i → ii



35) Choose the correct option with respect to stages shown by (A) and (B)

	A	B
(1)	Early Prophase	Anaphase
(2)	Late prophase	Anaphase
(3)	Late prophase	Metaphase-I
(4)	Early prophase	Telophase

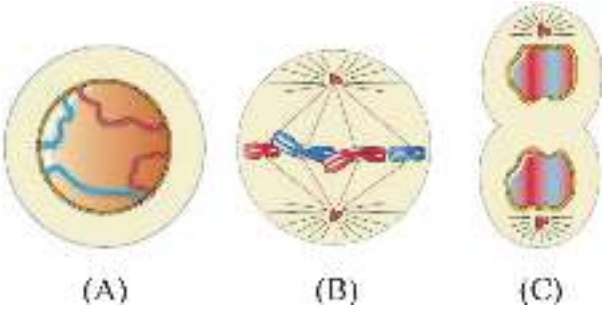
(1) 1

(2) 2

(3) 3

(4) 4

36) The figures below show three stages (A, B, C) of mitotic division. Select the option giving correct identification together with what it represents ?



Options :-

(1) B-Individual chromosomes can no longer be seen

(2) A-Initiation of condensation of chromosomal material

(3) C-Initiation of assembly of mitotic spindle

(4) B-Chromosomes consist of one chromatid

37) A root tip cells has 14 chromosomes, how many chromosomes will the cell have, after splitting of centromere in mitotic anaphase ?

(1) 14

(2) 28

(3) 7

(4) 21

38) In which stage of cell cycle protein synthesis takes place ?

(1) G_1-Phase

(2) G_2-Phase

(3) S-Phase

(4) All of these

39) Select the correct sequence of events occurs during prophase-I of meiotic division.
(a) Formation of synaptonemal complex.
(b) Formation chiasmata.
(c) Appearance of recombination nodules.
(d) Compaction of chromosomes starts.
(e) Terminalisation of chiasmata.
Choose the correct answer from the options given below :

(1) (a), (b), (c), (d), (e)

(2) (d), (a), (c), (b), (e)

(3) (b), (a), (d), (c), (e)

(4) (e), (b), (d), (c), (a)

40) **Statement-I :** Cell growth is a continous process during cell cycle.
Statement-II : DNA synthesis occurs only in s-phase of cell cycle.

(1) Both **Statement I** and **Statement II** are correct.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are incorrect.

41) Which one of the following stage of meiosis begins with the spliting of centromere?

(1) Anaphase

(2) Metaphase II

(3) Anaphase II

(4) Telophase II

42) Given below are two statements :
Statement-I : The complete disintegration of the nuclear envelop marks the start of Anaphase.
Statement-II : Meiosis involves two sequential cycles of nuclear and cell division called Meiosis I and Meiosis II but only a single cycle of DNA replication.
In the light of above statements, choose the correct answer from the options given below :

(1) **Statement I** and **Statement II** both true

(2) **Statement I** is true and **Statement II** is false

(3) **Statement I** is false and **Statement II** is true

(4) **Statement I** and **Statement II** both false

43) Match List-I with List-II :

List-I		List-II	
A.	Leptotene	I.	Terminalisation of chiasmata
B.	Zygotene	II.	Crossing over
C.	Pachytene	III.	Dissolution of the synaptonemal complex
D.	Diplotene	IV.	Bouquet stage
E.	Diakinesis	V.	Formation of synaptonemal complex

(1) A-IV, B-II, C-V, D-III, E-I

(2) A-IV, B-V, C-II, D-III, E-I

(3) A-IV, B-V, C-III, D-II, E-I

(4) A-IV, B-V, C-II, D-I, E-III

44) At which substage of prophase-I in meiosis, the four chromatids of each bivalent chromosomes becomes disinfect and clearly appears as tetrads?

(1) Pachytene

(2) Zygotene

(3) Diplotene

(4) Diakinesis

45) In an onion root tip cell number of chromosomes and DNA content at G_1 stage are 16 and 2C respectively. After s-phase number of chromosomes and DNA content respectively will be :

(1) 16, 2C

(2) 16, 4C

(3) 4C, 16

(4) 32, 4C

46) About 1200 ml of air is always remain inside the human lungs. It is described as :-

(1) Tidal volume

(2) Residual volume

(3) Expiratory reserve volume

(4) Inspiratory reserve volume

47) Oxygen dissociation curve shifts to right when :-

(1) pO_2 increases

(2) pCO_2 decreases

(3) pH decreases

(4) Temperature decreases

48)

Read the following statements :-
(a) Respiratory rhythm centre is present in medulla region
(b) Chemosensitive area is highly sensitive to CO_2 and hydrogen ions.

(c) The role of oxygen in the regulation of respiratory rhythm is quite insignificant (d) Pneumotaxic centre is present in pons region.
(e) Receptors associated with aortic arch and carotid artery can recognise changes in CO_2 and $H^{\oplus}$ concentration. How many statements are correct about regulation of respiration ?

(1) Three

(2) Four

(3) Two

(4) Five

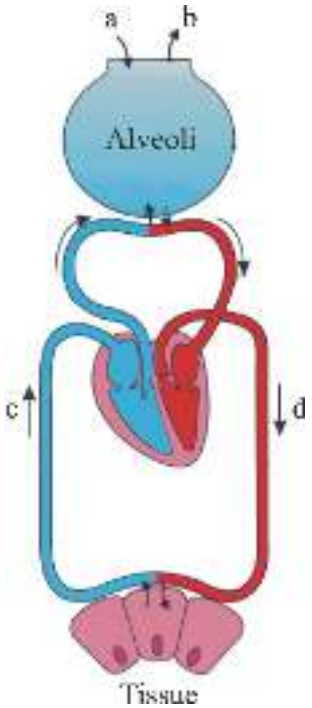
49) What would happen when blood is acidic ?

(1) Binding of oxygen with haemoglobin increases

(2) White blood cells are formed in higher number

(3) Binding of oxygen with haemoglobin decreases

(4) There is no change in oxygen binding nor number of RBC



50) Identify a, b, c and d in the given diagram.

	a	**b**	**c**	**d**
(1)	Inspired Air	Expired Air	More CO_2	More O_2
(2)	Inspired Air	Expired Air	More O_2	More CO_2
(3)	Expired Air	Inspired Air	More O_2	More CO_2
(4)	Expired Air	Inspired Air	More CO_2	More O_2

(1) 1

(2) 2
(3) 3
(4) 4

51) What will be the P_{O_2} and P_{CO_2} in the atmospheric air compared to those in the alveolar air ?

(1) P_{O_2} lesser, P_{CO_2} higher
(2) P_{O_2} higher, P_{CO_2} lesser
(3) P_{O_2} higher, P_{CO_2} higher
(4) P_{O_2} lesser, P_{CO_2} lesser

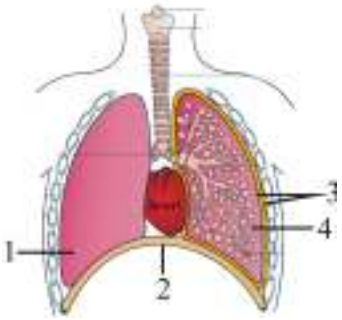
52) How many cellular layers are present in the diffusion membrane in lungs ?

(1) one
(2) two
(3) three
(4) four

53) If total O_2 transported by blood from lungs to cell in a minute is 500 ml. Then the volume of O_2 transported by blood found dissolved in plasma is

(1) 20 ml
(2) 15 ml
(3) 5 ml
(4) 30 ml

54) The given figure shows the respiratory system. Identify the correct structure marked as 1,2,3 and 4 whose contraction initiated the inspiration which in turn increases the volume of thoracic chamber in the antero–posterior axis :-



(1) 1–Lung
(2) 2–Diaphram
(3) 3–Pleural membranes
(4) 4–Alveoli

55) Proliferation of fibrous tissue in lungs of stone mine workers is characterstic feature of :-

(1) Asthma

(2) Occupational Respiratory Disorder

(3) Emphysema

(4) Bronchitis

56) Which respiratory capacity is correctly matched with its formula & value :-

	Respiratory capacities	**Formula**	**Value**
(1)	Functional residual capacity	ERV + RV	2300 ml
(2)	Vital capacity	ERV+RV+IRV	4600 ml
(3)	Total lung capacity	RV+IRV+ERV	5800 ml
(4)	Inspiratory capacity	TV+IRV	1600 ml

(1) 1

(2) 2

(3) 3

(4) 4

57) Functional residual capacity (FRC) is :

(1) TV + ERV

(2) TV + ERV + IRV

(3) ERV + RV

(4) TV + ERV + IRV + RV

58) A chemosensitive area is highly sensitive to :-

(1) O_2 and H^+ ions

(2) CO_2 and Na^+ ions

(3) CO_2 and H^+ ions

(4) CO_2 and O_2

59) How many of the following statements are true ?
(i) Mechanism of breathing in different organism depend on their habitat and level of organisation.
(ii) Gills are used by aquatic Arthropods.
(iii) Pharynx is a common pathway for food and air.
(iv) The initial bronchioles are supported by incomplete cartilaginous rings.

(1) Four

(2) Three

(3) Two

(4) One

60) The partial pressure of CO_2 in atmospheric air, oxygenated blood and deoxygenated blood respectively is :- (in mm of Hg)

(1) 104, 45, 40

(2) 95, 40, 45

(3) 0.3, 40, 45

(4) 0.3, 45, 40

61) Total lung capacity is :-

(1) total volume of air accommodated in lungs at the end of forced inspiration

(2) RV + ERV + TV + IRV

(3) vital capacity + residual volume

(4) All of the above

62) The conditions which are favourable for the formation of oxyhaemoglobin :

(1) PO_2 ↑, PCO_2 ↑, H^+ conc.↓, Temperature↓

(2) PO_2↓, PCO_2↓, H^+ conc. ↑, Temperature ↑

(3) PO_2↑, PCO_2↓, H^+ conc.↓, Temperature↓

(4) PO_2 ↑, PCO_2↑, H^+ conc.↑, Temperature ↑

63) A person breathes in volume of air by forced inspiration after having a forced expiration. This quantity of air taken in is

(1) Total lung capacity

(2) Tidal volume

(3) Vital capacity

(4) Inspiratory capacity

64)

How many animals in the given list have gills for respiration?
Earthworm, Frog, Fish, Insects, Reptiles, Aquatic arthropods, Flatworm, Molluscs, Birds

(1) Six

(2) Four

(3) Three

(4) Five

65)

In normal condition Every 100 ml of oxygenated blood can deliver approximately :-

(1) 5 ml of O_2 to tissue

(2) 10 ml of O_2 to tissue

(3) 15 ml of O_2 to heart

(4) 20 ml of O_2 to tissue

66) Which of the following is not **correct** match :-

(1) Haemoglobin oxygen dissociation curve - sigmoid
(2) Carbamino haemoglobin - 97% (CO_2)
(3) Pneumotaxic centre - pons
(4) Emphysema - alveolar walls are damaged

67) Epiglottis is made up of:

(1) Elastic cartilage
(2) Sesamoid bone
(3) Hyaline cartilage
(4) Fibrous cartilage

68) Conducting part of the human respiratory system comprises of

(1) External nostrils upto the terminal bronchioles
(2) Internal nostrils upto trachea
(3) Epiglottis upto trachea
(4) Larynx upto bronchi

69) **Assertion (A) :** The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the pulmonary cavity.
Reason (R) : Such an arrangement is essential for breathing as we cannot directly alter the pulmonary volume.

(1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(2) Assertion are correct True but Reason is not correct.
(3) Assertion is not correct but reason is correct.
(4) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

70) **Assertion :** Larynx is also called sound box.
Reason : It is a cartilaginous box which helps in sound production.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

71) **Assertion (A) :-** The branching network of bronchi ,bronchioles and alveoli comprise the lungs.
Reason (R) :- Human have two lungs which are covered by single layered pleura.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(2) **(A)** is correct but **(R)** is not correct
(3) **(A)** is incorrect but **(R)** is correct
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

72) **Statement-I:** About 97% O_2 and nearly 20-25% CO_2 is transported by haemoglobin.
Statement-II : Very high concentration of carbonic anhydrase present in plasma and its minute quantities present in RBC.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **Statement II** are correct.

73) **Statement-I :** We have the ability to increase the strength of inspiration and expiration with the help of additional muscles in the abdomen.
Statement-II : The volume of air involved in breathing movements can be estimated by using a spirometer.

(1) Both statement I and statement II are correct.
(2) Both statement I and statement II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement I is incorrect but statement II is correct.

74) Read the following statements carefully and answer accordingly.
Statement-I :- IRV is the additional volume of air, a person can inspire by a forcible inspiration this average about 2500-3000ml.
Statement-II :- IC includes tidal volume and IRV.

(1) Statement-I is correct, Statement-II is incorrect
(2) Statement-I is incorrect, Statement-II is correct
(3) Statement-I and Statement-II both are correct
(4) Statement-I & Statement-II both are incorrect

75) **Statement-I :** Trachea is a straight tube extending upto mid thoracic cavity which devides at 5th thoracic vertebra into left and right primary bronchioles.
Statement-II : The respiratory passage from trachea to terminal bronchioles are supported by complete cartilaginour ring.

(1) Both statement-I and II are true
(2) Both statement-I and II are false
(3) Statement-I is correct but II is false
(4) Statement I is incorrect but II is correct.

76) How many organism in the list given below do respiration by general body surface.
Insects, Sponges, Flatworm, Fishes, Birds, Coeleterates, Reptiles

(1) 3
(2) 4
(3) 5
(4) 6

77) Match the following and mark the correct option :

Animal		Respiratory organ	
A.	Fish	(i)	Lungs
B.	Birds	(ii)	Tracheal tubes
C.	Insect	(iii)	Moist cuticle
D.	Earthworm	(iv)	Gills

(1) A-i, B-ii, C-iii, D-iv

(2) A-iv, B-i, C-ii, D-iii

(3) A-iv, B-i, C-iii, D-ii

(4) A-i, B-iv, C-iii, D-ii

78) Diaphragm increases thoracic volume in what axis ?

(1) Antero - Posterior Axis

(2) Dorso – Ventral Axis

(3) Ventro - Dorsal Axis

(4) Lateral axis

79) Which statement is incorrect regarding thoracic chamber ?

(1) In this chamber lungs are situated

(2) This chamber is an air tight chamber

(3) This chamber dorsally formed by vertebral column & ventrally by sternum and laterally by ribs and lower side by dome shaped diaphragm.

(4) This chamber dorsally formed by sternum, ventrally by vertebral column and laterally by Ribs and lower side by dome shaped diaphragm.

80) Arrange the given steps of respiration mechanism in the order they occur in human body :
(I) Breathing or pulmonary ventilation.
(II) Diffusion across the alveolar membrane.
(III) Transport of gases by blood.
(IV) Utilization of O_2 by cells.
(V) Diffusion of O_2 and CO_2 between blood and tissue.

(1) I → II → III → IV → V

(2) I → II → III → V → IV

(3) I → III → II → V → IV

(4) I → III → II → IV → V

81) Read the following terminology of column I & column II and identify their correct match :-

Column-I		Column-II	
A.	Annelida	i	Open Circulatory

B.	Arthopoda	ii	Closed Circulatory
C.	Mollusca	iii	Closed Circulatory
D.	Chordates	iv	Open Circulatory

(1) A - (i) , B - (ii) , C - (iii) , D - (iv)

(2) A - (iv) , B - (iii) , C - (ii) , D - (i)

(3) A - (ii) , B - (i) , C - (iv) , D - (iii)

(4) A - (iv) , B - (ii) , C - (iii) , D - (i)

82) Read the column-I, II & III and identify their correct match -

Column-I		**Column-II**		**Column-III**	
(A)	A blood group	(P)	A, B antigen	(i)	Anti-B antibodies
(B)	B blood group	(Q)	A antigen	(ii)	No antibodies
(C)	O blood group	(R)	B antigen	(iii)	Anti-A antibodies
(D)	AB blood group	(S)	No antigen	(iv)	Anti-A, B antibodies

(1) (A)-(P)-(i), (B)-(Q)-(iv), (C)-(R)-(ii), (D)-(S)-(iii)

(2) (A)-(Q)-(i), (B)-(R)-(iii), (C)-(S)-(iv), (D)-(P)-(ii)

(3) (A)-(S)-(ii), (B)-(P)-(iv), (C)-(Q)-(i), (D)-(R)-(iii)

(4) (A)-(R)-(iii), (B)-(S)-(i), (C)-(P)-(ii), (D)-(Q)-(iv)

83) Read following terminology of column I & II and identify their correct match

	Column-I		**Column-II**
(A)	Sino-atrial node	(i)	Between left atrium and left ventricle
(B)	Atrio-ventricular node	(ii)	Between right atrium and right ventricle
(C)	Tricuspid valve	(iii)	In upper right corner of right atrium
(D)	Bicuspid valve	(iv)	In lower left corner of right atrium

(1) A-(iv), B-(iii), C-(ii), D-(i)

(2) A-(i), B-(ii), C-(iv), D-(iii)

(3) A-(iii), B-(iv), C-(ii), D-(i)

(4) A-(ii), B-(i), C-(iii), D-(iv)

84) Read the column-I, II and III and identify their correct match -

Column-I		**Column-II**		**Column-III**	
(A)	RBC	(P)	Leucocytes	(I)	5-5.5 million mm^{-3}
(B)	WBC	(Q)	Thrombocytes	(II)	150000-350000 mm^{-3}

(C)	Platelets	(R)	Erythrocytes	(III)	6000-8000 mm^{-3}

(1) (A)-(R)-(I), (B)-(Q)-(II), (C)-(P)-(III)

(2) (A)-(Q)-(III), (B)-(R)-(II), (C)-(P)-(I)

(3) (A)-(P)-(III), (B)-(R)-(I), (C)-(Q)-(II)

(4) (A)-(R)-(I), (B)-(P)-(III), (C)-(Q)-(II)

85) **Assertion :** Leucocytes are also known as white blood cells.
Reason : They are colourless duc to lack of haemoglobin.

(1) Assertion & Reason both are correct but reason is correct explanation of Assertion.

(2) Assertion & Reason both are correct but reason is not correct explanation of assertion.

(3) Assertion is correct but reason is false.

(4) Assertion is false but reason is correct.

86) **Statement-I :** RBC's are formed in the red bone marrow.
Statement-II : RBC's have average life span of 120 days.

(1) Statement-I is correct but statement-II is incorrect.

(2) Statement-I is incorrect but statement-II is correct.

(3) Statement-I & statement-II both are incorrect.

(4) Statement-I & statement-II both are correct.

87) Choose the correct pathway of the transmission of impulses in the heart beat :

(1) AV node → S A node → Bundle of His → Purkinje fibres

(2) SA node → AV node → Bundle of His → Purkinje fibres

(3) SA node → Bundle of His → AV node → purkinje fibres

(4) AV node → Bundle of His → SA node → Purkinje fibres

88) Which of the following , does not help in clotting of blood ?

(1) Heparin

(2) Prothrombin

(3) Ca^{2+}

(4) Fibrinogen

89) How many statements is/are incorrect ?
(A) Blood is a special epithelial tissue
(B) Albumins are needed for clotting
(C) Plasma with the clotting factors is called serum
(D) Plasma is a straw coloured, viscous fluid

(1) One

(2) Two

(3) Three

(4) Four

90) Least number of white blood corpuscles are :-

(1) Neutrophils
(2) Basophils
(3) Monocytes
(4) Eosinophils

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	2	1	2	4	1	1	2	4	3	2	1	2	1	1	2	3	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	1	4	4	2	3	2	3	1	3	2	1	3	3	3	4	3	4	1
Q.	41	42	43	44	45															
A.	4	3	3	1	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	1	3	3	3	2	3	3	3	1	4	4	4	1	4	2	2	2	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	2	2	3	2	2	3	4	1	1	4	2	4	4	3	4	2	2	4
Q.	86	87	88	89	90															
A.	4	3	2	1	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	3	2	2	4	2	3	3	2	2	1	1	2	2	4	4	4	2	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	1	2	4	3	4	4	3	3	2	4	3	2	2	2	2	4	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	2	1	2	2	3	4	3	1	2	2	2	2	2	1	3	3	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	3	3	1	2	1	1	4	1	2	2	1	3	2	1	2	1	4	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	3	4	1	4	2	1	3	2										

SOLUTIONS

PHYSICS

1)

$y = (0.996)^{1/4} = (1 - 0.004)^{1/4}$
$= 1 - 0.001 = 0.999$

2)

3 cm, 4 cm, 12cm

3)

$(1 - \cos^2 15°) + (1 - \sin^2 15°)$
$= 2 - (\cos^2 15° + \sin^2 15°) = 2 - 1 = 1$

4)

$y = x^{-3/2}$
$\frac{dy}{dx} = -\frac{3}{2}x^{-5/2}$

5)

$4° = \left(4 \times \frac{\pi}{180}\right)$ radian $= \frac{\pi}{45}$ radian
$\sin(4°) = \sin\left(\frac{\pi}{45}\right) \cong \frac{\pi}{45}$

6) $y = mx + c$
$y = (\tan 135°)x + 2$
$\boxed{x = -y + 2}$

7)

$y = \ln(\cos x)$
$\frac{dy}{dx} = \frac{1}{\cos x} x (-\sin x)$
$\frac{dy}{dx} = -\tan x$
at $x = \frac{3\pi}{4}$
$\frac{dy}{dx} = -\tan\frac{3\pi}{4}$
$= - (-1) = +1$

8) $y = 2x^2 - 8x$, $\frac{dy}{dx} = 4x - 8$

$\frac{dy}{dx} = 0$

$\Rightarrow 4x - 8 = 0$

$\Rightarrow \boxed{x = 2}$

Now $\frac{d^2y}{dx^2} = 4 > 0$ [positive]

$\Rightarrow y = y_{mini}$ at $x = 2$,

So $y_{mini} = 2 \times 2^2 - 8 \times 2$

$= 8 - 16$

$= -8$

9) $\int \left(\frac{a}{x^2} + \frac{b}{x}\right) dx$

$a \int \frac{1}{x^2} dx + b \int \frac{dx}{x}$

$a\left(-\frac{1}{x}\right) + b \log x + C$

10)

$y = \sin(e^x)$

$\frac{dy}{dx} = \cos e^x \times e^x = e^x \cos e^x$

11)

$A = 5t^2 + 4t + 8$

$\frac{dA}{dt} = 10t + 4$

$t = 3$ sec

$\frac{dA}{dt} = 10 \times 3 + 4 = 34$

12)

$y = x^2 e^x$

$\frac{dy}{dx} = x^2 e^x + e^x 2x$

13) $\int_{-\frac{\pi}{2}}^{0} \sin x dx + \int_{0}^{\frac{\pi}{2}} \sin x dx$

$= [-\cos x]_{-\frac{\pi}{2}}^{0} + [-\cos x]_{0}^{\frac{\pi}{2}}$

$= -1 + 1 = 0$

14) $\int_3^5 \frac{1}{2x+3} = \frac{1}{2}[\ln(2x+3)]_3^5$

$= \frac{1}{2}\ln\left(\frac{13}{9}\right)$

15) $\int_0^{\pi/2} \sin x dx = [-\cos x]_0^{\pi/2}$

= – (0 – 1) = 1 unit

16) $\int_0^{2\pi} \cos^2\theta \, d\theta = \int_0^{2\pi} \left(\frac{\cos 2\theta + 1}{2}\right) d\theta$

$= \frac{1}{2}\left[\frac{\sin 2\theta}{2} + \theta\right]_0^{2\pi} = \pi$

17)

$S = \frac{a}{1-r} = \frac{1}{1-1/3} = \frac{3}{2}$

18) Area under F-t curve gives change in momentum (ΔP)

$\Delta P = \int F dt$

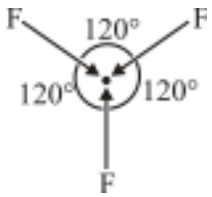
$\Delta P = \frac{1}{2} \times 6 \times 2 + (-3)(4-2) + 3 \times (8-4)$

ΔP = 6 – 6 + 12

ΔP = 12 Ns

19) $F = \frac{Nmv}{t} = \frac{5 \times 50 \times 10^{-3} \times 300}{1}$

F = 75 N



20)

Net Force

$F_{net} = 0$

$a_{net} = 0$

21) $F_{avg} = \frac{2mv\sin\theta}{t}$

$= \frac{2 \times 3 \times 10 \sin 60^{o}}{0.2} = 150\sqrt{3}N$

22) mass m = $\frac{|\vec{F}|}{|\vec{a}|} = \frac{\sqrt{200}}{1} = 10\sqrt{2}$

23) $F = m\left(\frac{dv}{dt}\right) = \frac{100 \times 5}{0.1} = 5000\,N$

24)

Explanation Question: We have to choose the incorrect option
Concept: This question is based on vectors
Solution:

$\left|\vec{A}\right| = \sqrt{3^2 + 4^2} = 5$

$\left|\vec{B}\right| = \sqrt{6^2 + 8^2} = 10$

$\vec{A}.\vec{B} = \left(3\hat{i} + 4\hat{j}\right).\left(6\hat{i} + 8\hat{j}\right)$

= 18 + 32 = 50

$\frac{\left|\vec{A}\right|}{\left|\vec{B}\right|} = \frac{5}{10} = \frac{1}{2}$

Final Answer: Option (4)

25) **Explanation** : resultant of which pair given in options cannot given the magnitude of 4N
Concept : Addition and subtraction
Solution : Max. Resultant = A + B
Mini. Resultant = |A – B|
$(A + B) \leqslant R \leqslant (|A - B|)$
A. $5 \leqslant R \leqslant 1$
B. $6 \leqslant R \leqslant 0$
C. $8 \leqslant R \leqslant 4$
D. $11 \leqslant R \leqslant 5$ 4 cannot be resultant in this addition
Final answer: D

26) **Explanation:**
Two vectors are given. The component of vector b perpendicular to vector a is zero. We have to find the value of x.
Concept:
This question is based on the condition of parallel vectors .
Solution:
Given vectors in vector form:
$\vec{a} = 3\hat{i} + 8\hat{j} - 2\hat{k}$
$\vec{b} = 6\hat{i} + 16\hat{j} + x\hat{k}$

If the component of $\vec{b}$ perpendicular to $\vec{a}$ is zero, then
$\vec{b} \parallel \vec{a}$
So,
$\vec{b} = \lambda \vec{a}$
Comparing components:
6 / 3 = 2
16 / 8 = 2
Hence λ = 2
From k-component:
x / (−2) = 2
x = −4
Final Answer: Option (2)

27) $\vec{S} = \vec{F}_1 + \vec{F}_2$ and $\vec{D} = \vec{F}_1 - \vec{F}_2$
As two vectors are perpendicular to each other, hence $\vec{S} \cdot \vec{D} = 0$.
or $\left(\vec{F}_1 + \vec{F}_2\right) \cdot \left(\vec{F}_1 - \vec{F}_2\right) = 0$
or $\left|\vec{F}_1\right|^2 - \left|\vec{F}_2\right|^2 = 0$
or $\left|\vec{F}_1\right|^2 = \left|\vec{F}_2\right|^2$
or $\left|\vec{F}_1\right| = \left|\vec{F}_2\right|$

28)

Explain Question : In this question, we are asked to find the velocity vector of a particle moving with speed 6m/s along the direction of $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$.
Concept : This question is based on vector in a given direction (unit vector)
Solution :-The velocity vector $\vec{V}$ in a given direction is obtained by multiplying the unit vector in that direction by the speed i.e. V.
$\vec{V} = V.\hat{A}$
where $\hat{A} = \dfrac{\vec{A}}{|\vec{A}|}$
Here $|\vec{A}| = \sqrt{(2)^2 + (2)^2 + (-1)}$
$|\vec{A}| = \sqrt{(2)^2 + (2)^2 + (-1)^2}$
$|\vec{A}| = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$
$\hat{A} = \dfrac{\vec{A}}{|\vec{A}|} = \dfrac{2\hat{i} + 2\hat{j} - \hat{k}}{3}$
So, $\vec{V} = v.\hat{A} \Rightarrow 6 \times \dfrac{(2\hat{i} + 2\hat{j} - \hat{k})}{3}$
$\vec{V} = 4\hat{i} + 4\hat{j} - 2\hat{k}$
Final answer : Option (2)

29) **Explain Question:**
In this question, we have to find the value of 'b' if $\vec{A} = 0.3\hat{i} + b\hat{j} - 0.4\hat{k}$ is a unit vector.

Concept:
Unit vector

Solution:
The magnitude of a unit vector is 1.

$\therefore \left|\vec{A}\right| = 1$

$\Rightarrow \sqrt{(0.3)^2 + (b)^2 + (-0.4)^2} = 1$

$\Rightarrow \sqrt{0.09 + b^2 + 0.16} = 1$

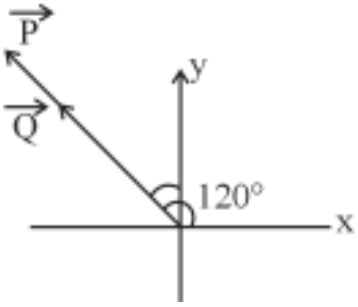
Squaring both side we get

$\Rightarrow b^2 + 0.25 = 1$

$\Rightarrow b^2 = 1 - 0.25$

$\Rightarrow b = \pm\sqrt{0.75}$

Final Answer: Option (3)



30)

Angle between $\vec{P}$ & $\vec{Q}$ is 0°

$\left|\vec{R}\right| = P + Q$

31) **Explain Question:**
The question asks for dot product of given vectors.

Concept:
Dot product of vectors.

Solution:

$(2\bar{A} + 3\bar{B}) \cdot (3\bar{A} - 2\bar{B})$

$=6A^2 - 4AB\cos\theta + 9\,BA\cos\theta - 6B^2$

$=6A^2 + 5\,AB\cos\theta - 6B^2$

Final answer: option (3)

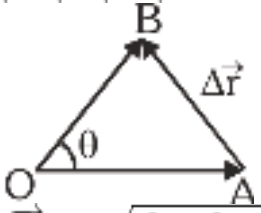
32) **Explain Question:** find how much the tip of vector shifts when the vector is rotated by angle about its tail.

Concept: Subtraction of vectors.

Solution:

$\overrightarrow{\Delta r} = \overrightarrow{OB} - \overrightarrow{OA}$

$|\overrightarrow{OA}| = |\overrightarrow{OB}| = \ell$



$|\overrightarrow{\Delta r}| = \sqrt{\ell^2 + \ell^2 - 2\ell^2\cos\theta}$

$= \sqrt{2\ell^2(1-\cos\theta)}$

$= \sqrt{2\ell^2 \times 2\sin^2\theta/2}$

$= 2\sin\theta/2$

Final Answer: option (2)

33) **Explain Question:**
We have to find calculate unit vector

Concept:
Unit vector

Solution:

$\vec{R} = \vec{a} + \vec{b} + \vec{c} = \hat{i} + \hat{j} - \hat{k}$

$\left|\vec{R}\right| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

$\hat{R} = \dfrac{\vec{R}}{\left|\vec{R}\right|} = \dfrac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

$\dfrac{1}{\sqrt{3}}\left(\hat{i} + \hat{j} - \hat{k}\right)$

Final Answer: Option (1)

34) **Explain Question:**
We have to calculate value of $(\vec{a}_1 - \vec{a}_2) \cdot (2\vec{a}_1 + \vec{a}_2)$

Concept:
Dot product

Solution:

$|\vec{a}_1| = |\vec{a}_2| = 1$

$|\vec{a}_1 + \vec{a}_2|^2 = |\vec{a}_1|^2 + |\vec{a}_2|^2 + 2\vec{a}_1 \cdot \vec{a}_2$

$\left(\sqrt{3}\right)^2 = 1 + 1 + 2\vec{a}_1 \cdot \vec{a}_2$

$\vec{a}_1 \cdot \vec{a}_2 = \dfrac{1}{2}$

$(\vec{a}_1 - \vec{a}_2) \cdot (2\vec{a}_1 + \vec{a}_2) = 2a_1{}^2 - 2a_1 \cdot a_2 + a_1 a_2 - a_2^2$

$= 2 - \dfrac{1}{2} - 1$

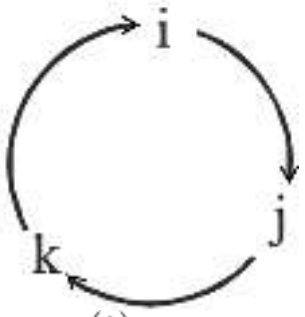
$= \frac{1}{2}$

Final Answer: Option (3)

35) **Question Explanation:** We need to find the value of $(2\hat{a} - 5\hat{b}).(3\hat{a} + \hat{b})$ from given data.
Concept : Application of dot product.
Solution :
$|\hat{a}| = |\hat{b}| = 1, |\hat{a} + \hat{b}| = \sqrt{3}$
$\Rightarrow |\hat{a}|^2 + |\hat{b}|^2 + 2|\hat{a}||\hat{b}|\cos\theta = 3$
$\Rightarrow \cos\theta = \frac{1}{2}$
$\Rightarrow \theta = 60°$(Angle between $\hat{a}$ and $\hat{b}$)
Now,
$(2\hat{a} - 5\hat{b}).(3\hat{a} + \hat{b})$
$= 6\hat{a}.\hat{a} - 15\hat{b}.\hat{a} + 2\hat{a}.\hat{b} - 5\hat{b}.\hat{b}$
$= 1 - 13\,\hat{a}.\hat{b}$
$= 1 - 13\,(1.1.\cos 60°)$
$= 1 - \frac{13}{2} = -\frac{11}{2}$
Final answer : option (3)

36) $\vec{A} \times \vec{B} = \vec{R}$
$\vec{R} \perp \vec{A}$
$\vec{R} \perp \vec{B}$
$\vec{R} \perp (\vec{A} + \vec{B})$
So $(\hat{a} + \hat{b}) \cdot (\hat{a} \times \hat{b}) = 0$
$= (\hat{a} + \hat{b}) \cdot (\hat{a} - \hat{b})$
$= \hat{a} \cdot \hat{a} - \hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{a} - \hat{b} \cdot \hat{b}$
$= 1 - 1$
$= 0$

37) $\vec{A} \times \vec{B} = 50\hat{j} \times 20(-\hat{i})$



$= 100(\hat{k})$ = 100 up

38) $\vec{F}_1 = \frac{-a}{\sqrt{2}}\hat{i} + \frac{a}{\sqrt{2}}\hat{j}$, $\vec{F}_2 = a\hat{i}$, $\vec{F}_3 = -a\hat{j}$

39)

Explain:
Evaluate $(\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{A})$.

Concept:
dot & cross product

Solution:
$(\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{A})$
$\downarrow \vec{P}$ $\downarrow \vec{Q}$
Here, $|\vec{P}| = ABSin\theta$
$|\vec{Q}| = BASin\theta$
$\vec{P} \cdot \vec{Q} = PQCos\theta$
$= ABSin\theta\ BASin\theta\ Cos180°$
$= -A^2B^2Sin^2\theta$
Note $\vec{P}$ & $\vec{Q}$ are opposite vectors hence $\theta = 180°$, $cos180° = -1$

Final Answer : Option (4)

40)

Question Explanation : The question asks whether the cross product $\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A} + \vec{B}$ & $\vec{A} - \vec{B}$ & whether $\vec{A} \times \vec{B}$ & $\vec{B} \times \vec{A}$ are antiparallel & non coplanar.
Concept: Based on vector addition, subtraction & multiplication.
Solution:
Statement1→ $\vec{A} + \vec{B}$ & $\vec{A} - \vec{B}$ are in the plane of $\vec{A}$ & $\vec{B}$ according to triangular law of vector addition.
Direction of $\vec{A} \times \vec{B}$ is perpendicular to the plane of $\vec{A}$ & $\vec{B}$ according to Right land thumb rule.
Hence assertion is correct.
Statement2→ **Coplanarity:** By definition, **any two vectors** in three-dimensional space are always **coplanar** because a single plane can always be found that contains both of them.
Since $\vec{A} \times \vec{B}$ & $\vec{B} \times \vec{A}$ lie along the same line (just in opposite directions), they are inherently coplanar.
Final answer: option (1)

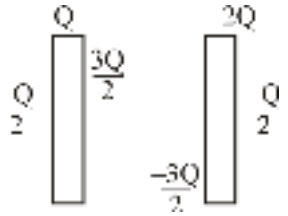
41) $n = \frac{Q}{e} = \frac{3.2 \times 10^{-20}}{1.6 \times 10^{-19}} = 0.2$
= where n should be integer

42)

Explaining the Question :- Two large metallic plates carry charges +Q and –2Q. Find

induced charge x on inner surface of left plate.
Concept:- This question is based on Charge distribution on inner and outer surfaces of parallel plates
Solution :-
Total charge of system = Q + (– 2Q) = – Q. This distributes equally on the two outermost surfaces, so each outer surface has – Q/2. Let inner charge on left plate be x. Then,

$$x + \left(-\frac{Q}{2}\right) = Q \Rightarrow x = Q + \frac{Q}{2} = \frac{3Q}{2}$$



The inner surface of the right plate must be equal and opposite, i.e. –3Q/2. This satisfies the total charge of –2Q on the right plate.
Final Answer :- 3

43) **1. Question Explanation:**
Two identical metallic spheres A and B carry charges of 7 μC and 1 μC respectively. When they are connected by a wire, charges redistribute due to conductivity and equal potential. We are to find the amount of charge that flows from A to B.
2. Concept Based:
Subtopic: Electrostatics - Redistribution of charge between conductors
When two identical conducting spheres are connected by a wire, charges redistribute such that both have the same potential. For identical spheres, this means equal charge on both.
3. Formula Used: $Q_{final} = \frac{Q_A + Q_B}{2}$ Charge that flows from A to B : $\Delta Q = Q_A - Q_{final}$ **4. Calculation:**
Initial charges:
$Q_A = 7\mu C$
$Q_B = 1\mu C$
Total charge: $Q_{total} = Q_A + Q_B = 7 + 1 = 8\mu C$
Final charge on each sphere (since identical): $Q_{final} = \frac{8}{2} = 4\mu C$
Charge flow from A to B : $\Delta Q = Q_A - Q_{final} = 7 - 4 = 3\mu C$

44) (A), (B), (C), (D) are correct statements.

45) **Asking about:**
electric force b/w two charges
Formula

$$F = \frac{kq_1q_2}{r^2}$$

Solution:
(i) $F_1 = \frac{k.(2)(6)}{r^2} = \frac{12k}{r^2} = 12N$
(ii) now – 4C is given to each of these charges

$$F_2 = \frac{k.(-2)(2)}{r^2} = \frac{-4k}{r^2}$$

$= (-4)(1N) = -4N$

$\Rightarrow F_2 = 4N$ attractive

Correct option: (2)

CHEMISTRY

46) Limiting reagent = (Moles ÷ Stoichiometric coefficient)

A: 5 ÷ 1 = 5

B: 8 ÷ 2 = 4 → smaller value → B is limiting reagent

Moles of C produced = moles of limiting reagent × (product coefficient ÷ reagent coefficient)

Moles of C produced = 8(1 ÷ 2)

Moles of C produced = 4 mole

47)

1. $\text{Molar mass}_{O_2} = 32g\,mol^{-1};\ \text{Molar mass}_{N_2} = 28g\,mol^{-1}$

2. Initial moles : $n_{O_2} = \frac{m}{32}; nN_2 = \frac{m}{28}$

3. Limiting reagent is the reactant with the smallest ratio of $\frac{\text{Initial moles}}{\text{Stoich. Coeff.}}$ (Stoich. Coeff. is 1 for both).

4. Comparison : Since $\frac{m}{32} < \frac{m}{28}, nO_2 < nN_2$

48)

Moles of $O_2 = \frac{32}{32} = 1$

$4A1 \times 3O_2 \rightarrow 2\ A1_2O_3$

$\frac{\text{moles of Al}}{4} = \frac{\text{moles } O_2}{3}$

Moles of Al = $\frac{4}{3} \times \text{moles of } O_2 = 4/3 \times 1 = 4/3$

Mass of Al reacted = $\frac{4}{3} \times 27 = 36g$

49) Reaction : $2H_2 + O_2 \rightarrow 2H_2O$

For H_2 : $\frac{4}{2} = 2$

For O_2 : $\frac{4}{1} = 4$

Smaller value = 2 → H_2 is the limiting reagent.

Now, from balanced equation :

2 moles H_2 → 2 moles H_2O

4 moles H_2 → 4 moles H_2O

Mass of water formed : 4 × 18 = 72 g

50)

1. $\text{Moles} = \frac{11.2\text{L}}{22.4\ \text{L/mol}} = 0.5\ \text{mol}$
2. Mass of pure $CaCO_3$:
   Ratio is 1 : 1, so 0.5 mol of $CaCO_3$ is needed.
   Mass = 0.5 mol × 100 g/mol = 50 g
3. Weight of 95% sample (W) :

% purity = $\frac{\text{Actual wt}}{\text{Total wt}} \times 100$

$95 = \frac{50}{\text{Total wt}} \times 100$

Total wt = 52.63 g

51)

Nitrogen forms several oxides: NO, N_2O, NO_2, N_2O_5.
Mass of oxygen combining with fixed mass of nitrogen varies in simple whole-number ratios.
This is a direct example of the law of multiple proportions.

52)

Based on theory

53) $2 \times 10^{-3} \times N_A \times 10$

54)

This question is based on
• At NTP, 1 mole of gas occupies 22.4 L
• Number of molecules = (Volume of gas / Molar volume) × Avogadro's number
Solution:
• Volume of O_2 = 1 L
• Number of molecules = $(1 / 22.4) \times 6.022 \times 10^{23}$

55) (I) 0.5 mole O_3 = 24 g O_3;
(II) 0.5 g atoms of oxygen = 8 g
(III) $\frac{3.011 \times 10^{23}}{6.022 \times 10^{23}} \times 32 = 16$ g O_2
(IV) $\frac{5.6}{22.4} \times 44$ g $CO_2 = 11$ g CO_2

56)

Let hydrocarbon be C_xH_y. Combustion:

$C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + y/2\ H_2O$
Given: 30 mL hydrocarbon reacts with 90 mL O_2, produces 60 mL CO_2
From CO_2: x = Volume CO_2 ÷ Volume hydrocarbon = 60 ÷ 30 = 2 → x = 2
O_2 required = x + y/4 = Volume O_2 ÷ Volume hydrocarbon = 90 ÷ 30 = 3
2 + y/4 = 3 → y/4 = 1 → y = 4
Molecular formula = C_2H_4

57) **Explain :** We have to find volume of water vapour.
Formula : PV = nRT
Calculation : 1 bar × V = $\frac{1.8}{18} \times 0.083 \times 650$
V = 5.37 L

58)

Law of gaseous volume.

59) **Explain the Question :**
We are asked to find the percentage composition of propane and butane in a 3L gas mixture based on CO_2 produced.
Concept : This question is based on combustion reaction stoichiometry and gas volumes.
Solution :
Combustion reactions :
$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\square)$
$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(\ell)$
Let volumes be : xLC_3H_8 and (3 – x) LC_4H_{10}
CO_2 produced : 3x + 4(3 – x) = 10
3x + 12 – 4x = 10 ⇒ –x + 12 = 10 ⇒ x = 2 L
So, C_3H_8 = 2L, C_4H_{10} = 1L
Percentage : $C_3H_8 = \frac{2}{3} \times 100 \approx 66.67\%$, 33.33%

60) $3O_2 \rightarrow 2O_3$
given vol. of O_2 = 90 mL
vol. of O_2 which is converting into O_3 is 10% of it = $\frac{90 \times 10}{100}$ = 9 mL.
After reaction (90 – 9) = 81 mL
$\frac{2}{3} \times 9$ = 6 mL O_3
vol. of the mixture of gases (O_2 & O_3)
= 81 + 6 = 87 mL
turpentine oil absorb O3, so after passing through turpentine oil vol. will be → 81 mL

61) 1 mol N^{-3} ion → 1 × 10 mol e^-
1 mol O^{-2} ion → 1 × 10 mol e^- = $6.023 \times 10^{24} e^-$
1 mol CH_4 ⇒ 1 × 10 mol protons

1 mol H_2O ⇒ 1 × 10 mol protons

62) NCERT Pg # 24
At of 1 atom of Al = $27 \times 1.66 \times 10^{-24}$ g
= 44.82×10^{-24} g

63)

$2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

 From 1 mole $KClO_3$ $\frac{3}{2}$ mole O_2 produced

64) **Concept:**
Avogadro's Law & Stoichiometry – 1 mole of any element contains 6.022×10^{23} atoms.
Solution:
Mass of Na_2CO_3 = 106 g, Molar mass = 106 g/mol → 1 mol → 1 mol C atoms → 6.022×10^{23} C atoms
Mass of C = 12 g, Molar mass = 12 g/mol → 1 mol → 6.022×10^{23} C atoms
Both have the same number of carbon atoms.
Reason (R) correctly explains Assertion (A) because both contain 1 g atom of carbon.

65)

In compound X_2O_3
 3 × 16 g of O is combined with
→ 2 × 91.5 g of X
 8g of O is combined with → $\frac{2 \times 91.5}{3 \times 16} \times 8$
= 30.5 g of X
So equivalent wieght of x = 30.5

66)

Theoretical mass of CO_2 from 150 g pure $CaCO_3$
(44 ÷ 100) × 150 = 66 g
Actual CO_2 evolved = 44g
Percentage purity = (44 ÷ 66) × 100 = 66.67%

67) In 28 g of N_2(g),
no. of atoms = $\frac{28}{28} \times N_A \times 2 = 2N_A$ atoms
In 32 g of O_2(g),
no. of atoms = $\frac{32}{32} \times N_A \times 2 = 2N_A$ atoms

68)

$M_{acid}V_{acid}n_{acid} = M_{base}V_{base}n_{base}$
$\Rightarrow (0.1\ M) \times (200\ mL) \times 1 = (0.5\ M) \times V_{base} \times 1.$
$20 = 0.5 \times V_{base}.$
$V_{base} = \frac{20}{0.5} = 40mL$

69) **Explain Question:** Check electronic configuration of ion belongs to which block

Concept: Electronic configuration

Solution:
$M^{+3} \rightarrow [Ar]3d^{10}$ (It is ion E.C.)
$M \rightarrow [Ar]\ 3d^{10}\ 4s^2\ 4p^1$ and it belongs to p-block.

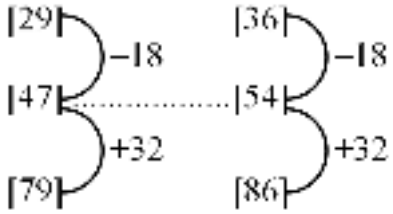
Final Answer: Option (2)

70) **Explain Question:** Identify general electronic configuration of d-block elements.

Concept: Electronic configuration

Solution: d-block elements involve filling of (n–1)d orbitals with 1–10 electrons and may have 0–2 electrons in ns due to stability shifts. Hence $ns^{0-2}\ (n-1)^{1-10}$ is correct .

Final Answer: Option (3)

71) 

72) **Explain Question:** Find the correct IUPAC symbol for Lv
Concept: IUPAC Name of elements
Solution: Last member of chalcogens is $_{116}Lv$
Z = 116 → ununihexium (Uuh)
Final Answer (2)

73) $A.R. \propto \frac{-ve O.S.}{+ve O.S.}$ for ions of same element & isoelectronic

74)

$Be^{+2} \rightarrow Be^{+3} + e^{\ominus}$ k^+, Na^+, Li^+
[He] [Ar] [Ne] [He]

75) **Question Explanation :**
Question is asking about Zeffective of Hydrogen
Concept : Zeffective of hydrogen.

Explanation & Solution :
The effective nuclear charge (Zeff) for a hydrogen atom is equal to its atomic number, which is 1. This is because a hydrogen atom has only one proton (the nucleus) and one electron, so there is no shielding of the electron by other electrons.

Answer : Option 1

76) **Explanation:-**

The question asks for the correct order of the shielding effect for different orbitals (s, p, d, f).

Concept:- Zeff

Solution:
More diffuse shape less shielding

S orbital = spherical shape
P orbital = dumble shape
D orbital = double dumble shape
F orbital = diffuse shape

Order of Shielding Effect:

A. s > p > d > f

Final Answer: Option (1). p > d > f

77)

$Li^+ \simeq Mg^{2+}$ (Diagonal relation)
Co ≃ Ni (3d-series)
Mo ≃ W (Lanthanide contraction)

78)

Answer - Option(2)
Explanation - Lanthanoid contraction significantly affects elements that come after the lanthanoids in the periodic table. This effect is seen most prominently in elements of groups where the corresponding 4d and 5d elements have nearly identical radii due to the poor shielding of f-electrons.
Concept - Lanthanoid contraction does not influence the comparison between Y and La because it primarily affects elements after the lanthanoids.

79)

Explain the Question : Identify the configuration that requires the least energy to remove an electron.
Concept : This question is based on Ionization Potential.
Solution : The configuration ending in $3s^2 3p^1$ has a single, shielded electron in the p-orbital. Removing it is easy and leaves a stable, filled $3s^2$ subshell, resulting in the lowest ionization potential.
In a common period I.P. order - $s^1 < p^1 < s^2 < p^2 < p^4 < p^3 < p^5 < p^6$
Final answer : 4

80)

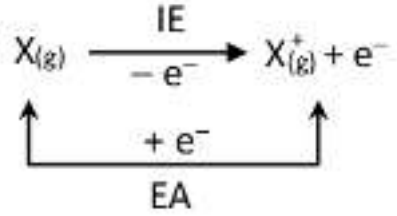
Explain the Question : Order elements by increasing second ionization energy.
Concept : This question is based on Ionization Energy.
Solution : After losing one electron, element B has a single p-electron, which is easiest to remove (lowest IE_2). And After losing one electron, Element D has a stable half-filled p-orbital (D^+ is ns^2np^3), making it hardest to remove from (highest IE_2).
$s^1 < p^1 < s^2 < p^2 < p^4 < p^3 < p^5 < p^6$
Final answer : 4

81) NCERT, Pg # 88, Fig.-3.5
Li Na K
Na is placed between Li & K in same group. So property will be intermediate.

82) In gaseous phase
Reducing action $\propto \frac{1}{IP}$

83) Cl > F > Br > I

84) **Explain Question :** The question relates the energy required to remove an electron from a neutral sodium atom (ionization energy) to the energy change when an electron is added to a sodium ion (electron gain enthalpy)
Concept : This Question is based on IP and EA
Solution :



Thus IE of $X_{(g)}$ = EA of $X^+_{(g)}$
So we can say IE of $Na_{(g)}$ = EA of $Na^+_{(g)}$
IE of $Na_{(g)} = 5.1eV$
$\therefore$ EA of $Na^+_{(g)} = -5.1eV$

Final Answer option (2)

85) $Cl + e^- \underset{\text{I.P. of } Cl^-}{\overset{\text{E.A. of Cl}}{\rightleftharpoons}} Cl^-$ [EA of Cl = I.P. of Cl^-]

86) Cl has the highest value of electron affinity in periodic table.

87)

Bond polarity ∝ ΔEN

- H-F is most polar bond

88)

It should be Br_2O
less EN element written first.

89) **Question Explanation:** Identify the substance that readily absorbs carbon dioxide gas.
Concept: This question is based on acid-base reaction
Solution: Carbon dioxide (CO_2) is an acidic oxide and is strongly absorbed by basic oxides.
K_2O is a strong basic oxide, while the others are amphoteric or acidic.
Final Answer: 1

90)

Acidic character $\propto$ EN of oxide and hydroxide
Acidic nature ∝ +ve O.S.

BIOLOGY

91)

NCERT XI Pg # 88

92)

NCERT XI Pg # 101

93)

NCERT XI Pg # 98

94) NCERT-XI, Pg. # 95

95) NCERT 11th Class : Cell Organelle

96)

NCERT XI Pg # 94

97) NCERT, Pg. # 99

98)

NCERT XI Pg # 99

99) NCERT, Pg. # 99

100) NCERT Pg # 100

101) NCERT Pg # 100

102) NCERT Pg # 99

103) NCERT Pg. # 96

104) NCERT Pg. # 97

105) NCERT Pg. # 97 - 98

106)

NCERT XI Pg # 93

107)

NCERT XI Pg # 94

108)

NCERT XI Pg # 94

109)

NCERT XI Pg # 94

110)

NCERT XI Pg # 94

111)

NCERT XI Pg. No. # 101

112)

NCERT XI Pg. No. # 95,96,98,99,100,101

113)

NCERT, Pg. # 99

114)

NCERT XI Pg # 94

115)

NCERT XI Pg # 98

116)

NCERT XI Pg # 97, 98

117)

NCERT XI Pg # 101

118)

NCERT XI Pg # 96, 97, 98

119) NCERT, Pg. # 97 & 98

120) NCERT, Pg. # 96 & 98

121) NCERT, Pg. # 97

122)

NCERT XI Pg # 121

123)

NCERT XI Pg # 121

124) NCERT (XI) Pg. # 122,123,124

125) **Question Explanation :**
The question asks to correctly identify the stages shown in both diagrams and select the option that accurately labels them.

Concept:
This question is based on phases of mitosis.

Explanation :
Diagram (A): The chromosomes are highly condensed, and the nuclear envelope is breaking down. This represents prophase. Since the chromosomes are very distinct and the nuclear envelope is clearly disintegrating, it is more specifically Late Prophase.

Diagram (B): The sister chromatids have separated and are moving to opposite poles of the cell. This is the characteristic event of Anaphase. The centromeres have split, and each chromatid is now considered a separate chromosome moving towards a pole.

Final answer: 2

126) NCERT XI (E/H) Pg # 165, 166

127) **Question Explanation :** The user is asking a question about the number of chromosomes in a root tip cell during a specific stage of mitosis.
Concept :
In mitosis, the number of chromosomes temporarily doubles at anaphase due to the splitting of centromeres.
• Before anaphase : 14 chromosomes (each with 2 sister chromatids)
• During anaphase : Centromeres split and each chromatid becomes an independent chromosome
• So the chromosome number becomes doubled.

Solution : Original cell : 14 chromosomes
After centromere splits in mitotic anaphase :
→ Each of the 14 chromosomes splits into 2 daughter chromosomes
→ 14 × 2 = 28 chromosomes (temperature, until cytokinesis)

Final answer : 2

128)

Question Explanation: Asking about S-Phase

Concept: S-Phase

Solution :

Protein synthesis occurs throughout interphase:

G1 phase → active protein synthesis for cell growth

S phase → proteins (like histones) are synthesized along with DNA replication

G2 phase → synthesis of proteins needed for mitosis

So, protein synthesis takes place in all these phases.

Final answer: 4

129) NCERT, Pg. # 126

130) NCERT, Pg. # 121

131) NCERT, Pg. # 127,128

132) NCERT Pg. # 123,125

133) NCERT-XI, Pg. # (E)-126, (H)-127

134) NCERT-XI, Pg. # (E)-126, (H)-126

135) NCERT-XI, Pg. # (E)-122, (H)-122

136)

Question Explanation : What is the volume of air that always remains inside the human lungs even after forceful expiration?

Concept: The question is based on lung volumes

Solution: The correct answer is option 2.

A. Residual volume is the amount of air (about 1100–1200 ml) that remains in the lungs after the most forceful expiration. This air prevents lung collapse and allows continuous gas exchange.

Final Answer: Option 2

137)

Explain question : To identify when oxygen dissociation curve shifts to right
Concept : This question is based on oxygen dissociation curve
Solution :- A rightward shift means hemoglobin releases oxygen more easily to tissues, Means Decreased affinity for oxygen.
This occurs at High CO_2 ($\uparrow CO_2$) Low pH (acidic), High temperature
Final answer :- 3

138)

Explain Question: Identify the number of correct statements about respiratory regulation.

Concept: Regulation of respiration

Solution: The correct answer is Option 4.

A. (a) True – Respiratory rhythm center is in the medulla oblongata.

B. (b) True – Chemosensitive area detects CO_2 and H^+ levels.

C. (c) True – Oxygen has a minor role; CO_2 is the main regulator.

D. (d) True – Pneumotaxic center is in the pons, regulating breathing rate.

E. (e) True – Aortic and carotid receptors sense CO_2 and H^+ concentration.

All five statements are correct.

Final answer: Option (4)

139)

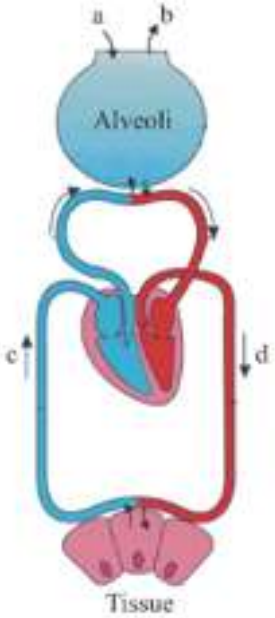
Explain Question : We are asked what happens to oxygen binding with hemoglobin when blood becomes acidic/high H^+ concentration.

Concept : This question is based on Transport of oxygen.

Solution : Acidic blood (low pH, high H^+ concentration) → reduces hemoglobin's affinity for oxygen → O_2 is released more easily to tissues.

Final Answer : option (3) Binding of oxygen with haemoglobin decreases

140) **Explain Question :** Identify a, b, c and d in the given diagram.
Concept : This question is based on transportation of gases
Explanation :



Final answer : 1 a = Inspired Air, b = Expired Air, c = More CO_2, d = More O_2.

141)

Question Explanation: The question asks how partial pressures of O_2 and CO_2 in atmospheric air compare to alveolar air.
Concept: Respiratory physiology – Partial pressures of gases
Solution: Atmospheric air has $PO_2 \approx 159$ mm Hg and $PCO_2 \approx 0.3$ mm Hg. Alveolar air has $PO_2 \approx 104$ mm Hg and $PCO_2 \approx 40$ mm Hg. Therefore, PO_2 in atmospheric air is higher than in alveolar air, and PCO_2 in atmospheric air is lower than in alveolar air.
Final Answer: Option 2

142) NCERT (XI) Pg. # 273

143)

Explain Question:
To identify volume of O_2 transported by blood, found dissolved in plasma, if total O_2 transported by blood from lungs to cell in a minute is 500 ml

Concept:
The concept based on O_2 transport

Solution:
The % of O_2 carried by plasma in dissolved form = 3%
so 3% of 500 ml

$$= \frac{3}{100} \times 500 = 15\text{ml}$$

Final Answer : (2)

144)

Explain the Question : The question asks to identify the structure whose contraction initiates inspiration and increases the thoracic chamber volume in the antero–posterior axis.
Concept : This question is based on mechanism of breathing.
Solution : The diaphragm contracts during inspiration, flattening downward, which increases the volume of the thoracic chamber in the antero–posterior axis. Other structures like lungs, pleural membranes, and alveoli do not actively initiate inspiration.
Final answer : 2

145)

Question Explanation:
The question asks about a lung condition caused by inhalation of dust in stone mines.

Concept: Disorders of respiratory system

Solution:

A. Occupational Respiratory Disorder → correct, includes silicosis

B. Silicosis is a type of occupational respiratory disorder caused by inhalation of silica dust.

C. It leads to proliferation of fibrous tissue (fibrosis) in the lungs, reducing lung elasticity.

Final answer: Option (2). Occupational Respiratory Disorder

146)

Explain Question : This question asks about which respiratory capacity is correctly matched with its formula & value?
Concept : This question is based on Respiratory volume and capacities
Explanation :

A. Functional Residual Capacity (FRC) = ERV + RV ≈ 2200–2300 ml

B. Vital Capacity (VC) = TV + IRV + ERV ≈ 4600 ml (not RV)

C. Total Lung Capacity (TLC) = VC + RV ≈ 5800 ml (not just RV+IRV+ERV)

D. Inspiratory Capacity (IC) = TV + IRV ≈ 3600 ml (not 1600 ml)

Only Functional Residual Capacity is correctly matched with its formula and value.
Final answer : 1

147)

Explain Question : This question asking about functional residual capacity (FRC) is
Concept : This question is based on Respiratory volume and capacities
Explanation : Functional residual capacity is the volume of air remaining in the lungs after normal expiration, and is equal to Expiratory Reserve Volume (ERV) + Residual Volume (RV).
Final answer : 3

148)

Explain Question : This question asking about a chemosensitive area is highly sensitive to
Concept : This question is based on Regulation of respiration
Explanation : The chemosensitive area near the medulla detects changes in CO_2 and H^+ ion concentration in body fluids, which strongly influence the rate of respiration.
Final answer : 3

149)

Question explanation: Count how many given statements about respiration and breathing are true.

Concept: Breathing and respiration – Respiratory organs & pathways

Solution:

(i) True - Mechanism of breathing varies with habitat and level of organization (e.g., skin in earthworm, gills in fish, lungs in mammals).

(ii) True - Aquatic arthropods (e.g., prawns, crabs) respire using gills.

(iii) True - Pharynx is a common passage for both food and air.

(iv) True - the trachea, primary, secondary and tertiary bronchi, and **initial bronchioles** are supported by incomplete cartilaginous rings.

Number of true statements = 4

Final answer: Option 1

150)

Explain Question : This question asking about the partial pressure of CO_2 in atmospheric air, oxygenated blood and deoxygenated blood respectively is
Concept : This question is based on exchange of Gases
Explanation :
Atmospheric air: $pCO_2 \approx 0.3$ mmHg
Oxygenated blood: $pCO_2 \approx 40$ mmHg
Deoxygenated blood: $pCO_2 \approx 45$ mmHg
So the sequence is $0.3 \rightarrow 40 \rightarrow 45$.
Final answer : 3

151)

NCERT XI, Pg. # 187

152)

Question Explanation : The conditions which are favourable for the formation of oxyhaemoglobin are:
Concept: Factors affecting oxygen-hemoglobin binding and the oxygen dissociation curve.
Solution:
Oxyhemoglobin (hemoglobin bound to oxygen) formation is favored under conditions of high oxygen partial pressure (PO_2), low carbon dioxide partial pressure (PCO_2), low H^+ concentration (high pH), and low temperature. These conditions are typically found in the lungs where oxygen uptake occurs.
High PO_2: In the lungs, the partial pressure of oxygen is high, favoring the binding of oxygen to hemoglobin.

A. **Low PCO_2 :** Low PCO_2 means less carbonic acid formation, hence lower H^+ concentration and higher pH.

B. **Low H^+ concentration (high pH):** A less acidic environment promotes oxygen binding to hemoglobin.

C. **Low Temperature :** Lower temperatures favor the stability of the oxyhemoglobin bond.

Final Answer: Option (3)

153)

Question explanation: The air inhaled after a forced expiration by maximum effort is the largest volume the lungs can move.

Concept: Respiratory volumes

Solution: The correct answer is option 3.

A. Vital capacity is the total volume of air a person can inhale after a forced exhalation, including tidal volume, inspiratory reserve, and expiratory reserve; it does not include residual volume.

Final answer: Option (3)

154)

Question Explanation: The question asks how many animals from the given list respire using **gills**.

Concept: The question is based o**n respiratory organs in different animals.**

Solution: The correct answer is option 3.

A. **Fish** respire through gills.
B. **Aquatic arthropods** (e.g., prawns, crabs) respire through gills.
C. **Molluscs** (aquatic forms) respire through gills.

A. Earthworm → skin respiration
B. Frog → lungs and skin (gills only in tadpole stage)
C. Insects → tracheal system
D. Reptiles → lungs
E. Flatworm → diffusion
F. Birds → lungs

So, total 3 among the given list respire through gills

Final answer: Option 3

155)

Explain the Question : O_2 delivered per 100 ml blood under normal conditions.
Concept : This question is based on transport of oxygen.
Solution : Each 100 ml of oxygenated blood carries ~ 20 ml O_2.
Under normal physiological conditions, about 25% of this oxygen is released to tissues.
25% of 20 ml = 5 ml O_2 delivered per 100 ml blood.
Thus, normally 5 ml O_2 is delivered to tissues from every 100 ml of oxygenated blood.
Final answer : 1

156)

Question Explanation: The question asks to identify the incorrectly matched pair related to respiratory physiology and blood components.

Concept: The question is based on breathing and exchange of gases.

Solution: The incorrect match is option 2.

A. Carbaminohemoglobin represents CO_2 bound to hemoglobin, but it accounts for only about 20-25% of CO_2 transport, not 97%.

Final answer: Option 2

157)

NCERT-XIth, Pg. # 184

158)

Explain Question : Conducting part of the human respiratory system includes.

Concept : This question is based on Respiratory tract.

Solution : Conducting part of the human respiratory system extends from external nostrils upto the terminal bronchioles.

Final Answer : option (1)

159)

Question explanation: Evaluate the correctness of the statements about **lung–thorax relationship during breathing**.
Concept: Respiratory system - lungs and thoracic cage
Solution: The correct answer is option 4.

A. The **assertion is true** because the lungs are enclosed within the thoracic cavity, so any change in thoracic volume is directly reflected in pulmonary volume.
B. The **reason is also true and correctly explains the assertion**, as lungs lack muscles to change their volume directly, making this anatomical arrangement essential for breathing.

Final answer: Option 4

160) NCERT Page No. # 269

161)

Question Explanation
The question is asking to evaluate the correctness of the statements about lung structure and pleura.

Concept
Lung and alveoli.

Solution
The branching network of bronchi, bronchioles, and alveoli makes up the lungs — (A) is correct. Humans have two lungs, but each lung is covered by a double-layered pleura, not single-layered — (R) is incorrect.

Final answer: Option 2

162)

Explain Question : This question is asking about: transport of O_2, CO_2 and location of carbonic anhydrase.
Concept : This question is based on Transport of respiratory gases
Solution : About 97% O_2 and 20–25% CO_2 are transported by hemoglobin (true). Carbonic anhydrase is highly concentrated in RBCs, not plasma (so Statement II is false).
Final answer : 2

163)

NCERT Page No. 186

164) NCERT Pg. # 186, 187

165)

Question Explanation: Determine the correctness of the two statements about trachea and bronchi.

Concept: Structure of trachea and bronchial tree

Solution: The correct answer is option 2

Statement-I False

A. Trachea divides at the level of the 5th thoracic vertebra (T5) into right and left primary bronchi but it divides into primary bronchi, not primary bronchioles.

Statement-II False

A. Trachea has C-shaped (incomplete) cartilaginous rings, not complete rings.

B. Bronchi also have cartilaginous support, but it is not in the form of complete rings throughout up to terminal bronchioles.

Thus, both statements are incorrect.

Final Answer: Option 2

166) NCERT Pg. # 268, Para 2

167)

Question explanation: Match the animals with their correct respiratory organs.
Concept: Respiratory organs

Solution:

A. **Fish → Gills →** (iv)

B. **Birds → Lungs →** (i)

C. **Insect → Tracheal tubes** → (ii)

D. **Earthworm → Moist cuticle** → (iii)

Correct matching: A–iv, B–i, C–ii, D–iii

Final answer: Option 2

168)

Explain the Question : Diaphragm increases thoracic volume what axis.
Concept : This question is based on mechanism of breathing.
Solution : Diaphragm increases thoracic volume in Antero-Posterior Axis.
Final answer : 1

169) NCERT-XI, Pg # 270

170) NCERT Pg. # 185

171)

NCERT (XIth) Pg. # 197

172) NCERT Pg.#195.

173)

NCERT (XIth) Pg. # 198, 199

174) NCERT Pg.#194.

175) NCERT Pg.#194.

176) NCERT Pg.#279.

177) NCERT Pg.#282

178) Heparin is an anticoagulant, meaning it prevents blood clotting. The other options, prothrombin, Ca^{2+}, and exposure to O_2, all play a role in the blood clotting process.

179)

Question explanation: Identify how many statements about blood are incorrect.
Concept: Blood composition and function

Solution: The correct answer is **option 3 – Three**.

A. **(A) Blood is a special epithelial tissue → Incorrect**
Blood is a **connective tissue**, not epithelial.

B. **(B) Albumins are needed for clotting → Incorrect**
Fibrinogen and other clotting factors are needed for clotting, not albumin.

C. **(C) Plasma with the clotting factors is called serum → Incorrect**
Serum is plasma **without clotting factors**.

D. **(D) Plasma is a straw coloured, viscous fluid → Correct**

Final answer: Option 3

180) **Explain Question :** Least number of white blood corpuscles are
Concept : This question is based on types of WBCs
Explanation : Basophils are present in the lowest proportion (~0.5–1% of total WBC count).
Final answer : 2 – Basophils